

User Manual

for S32K1_S32M24X ETH_43_ENET Driver

Document Number: UM2ETH_43_ENETASRR21-11 Rev0000R3.0.0 Rev. 1.0

1 Revision History	2
2 Introduction	3
2.1 Supported Derivatives	3
2.2 Overview	3
2.3 About This Manual	4
2.4 Acronyms and Definitions	5
2.5 Reference List	5
3 Driver	7
3.1 Requirements	7
3.2 Driver Design Summary	7
3.3 Hardware Resources	8
3.4 Deviations from Requirements	8
3.5 Driver Limitations	23
3.6 Driver usage and configuration tips	23
3.6.1 External data buffer usage for transmission and reception	23
3.6.1.1 Code snippets for External Buffers feature	24
3.7 Runtime errors	26
3.7.1 Standardized Extended Production Errors	26
3.7.2 Vendor-Specific Extended Production Errors	26
3.8 Symbolic Names Disclaimer	26
3.9 Usage in non AUTOSAR context	26
4 Configuration Parameters	29
4.1 Eth	29
4.1.1 Module Eth	34
4.1.2 Container EthGeneral	34
4.1.3 Parameter EthDevErrorDetect	34
4.1.4 Parameter EthGetDropCountApi	35
4.1.5 Parameter EthGetEtherStatsApi	35
4.1.6 Parameter EthGetCounterValuesApi	36
4.1.7 Parameter EthGetRxStatsApi	36
4.1.8 Parameter EthGetTxStatsApi	37
4.1.9 Parameter EthGetTxErrorCounterValuesApi	37
4.1.10 Parameter EthSendMultiBufferFrameApi	37
4.1.11 Parameter EthGlobalTimeSupport	38
4.1.12 Parameter EthVersionInfoApi	38
4.1.13 Parameter EthIndex	39
4.1.14 Parameter EthMainFunctionPeriod	39
4.1.15 Parameter EthMaxCtrlsSupported	39
4.1.16 Parameter EthMaxBuffersInMultiBufferFrame	40
4.1.17 Reference EthEcucPartitionRef	40
4.1.18 Container EthCtrlOffloading	41
4.1.19 Parameter EthCtrlEnableOffloadChecksumICMP	41
4.1.20 Parameter EthCtrlEnableOffloadChecksumIPv4	42
4.1.21 Parameter EthCtrlEnableOffloadChecksumTCP	42
4.1.22 Parameter EthCtrlEnableOffloadChecksumUDP	42
4.1.23 Container EthGeneralVendorSpecific	43
4.1.24 Parameter EthDisableDemEventDetect	43
4.1.25 Parameter EthEnableUserModeSupport	43
4.1.26 Parameter EthMultiPartitionSupport	44
4.1.27 Parameter EthUpdatePhysAddrFilterApi	44
4.1.28 Parameter EthSwtManagementSupportApi	45

4.1.29	Parameter	EthCoalescingInterrupt	45
4.1.30	Parameter	EthTimeoutMethod	46
4.1.31	Parameter	EthTimeoutDuration	46
4.1.32	Parameter	EthEnableCacheManagement	47
4.1.33	Container	EthConfigSet	47
4.1.34	Container	EthCtrlConfig	47
4.1.35	Parameter	EthCtrlEnableMii	48
4.1.36	Parameter	EthCtrlEnableMmd	48
4.1.37	Parameter	EthCtrlEnableRxInterrupt	49
4.1.38	Parameter	EthCtrlEnableTxInterrupt	49
4.1.39	Parameter	EthCtrlIdx	49
4.1.40	Parameter	EthCtrlMacLayerType	50
4.1.41	Parameter	EthCtrlMacLayerSubType	50
4.1.42	Parameter	EthCtrlMacLayerSpeed	51
4.1.43	Parameter	EthCtrlPhyAddress	51
4.1.44	Parameter	EthCtrlEnableSpiInterface	52
4.1.45	Parameter	EthCtrlConfigSwBufferHandling	52
4.1.46	Reference	EthCtrlEcucPartitionRef	52
4.1.47	Container	EthCtrlVendorSpecific	53
4.1.48	Container	EthCtrlConfigGeneral	53
4.1.49	Parameter	EthCtrlEnableAtInit	54
4.1.50	Parameter	EthCtrlAllocateTxDataBuffers	54
4.1.51	Parameter	EthCtrlAllocateRxDataBuffers	55
4.1.52	Parameter	EthCtrlEthIfIdx	55
4.1.53	Parameter	EthEthTrcvDriverVendorIdAndApiInfix	56
4.1.54	Parameter	EthEthSwtDriverVendorIdAndApiInfix	56
4.1.55	Parameter	EthDuplexMode	56
4.1.56	Parameter	EthMaxFrameLength	57
4.1.57	Parameter	EthTimeStampRequiredAccuracy	57
4.1.58	Parameter	EthTimeStampCorrectionCounter	58
4.1.59	Reference	EthModuleReferenceClock	59
4.1.60	Reference	EthTimeStampReferenceClock	59
4.1.61	Container	EthCtrlConfigInterrupts	60
4.1.62	Parameter	EthCtrlConfigEventCallback	60
4.1.63	Parameter	EthCtrlConfigTimeStampCallback	60
4.1.64	Parameter	EthCtrlEnableErrorInterrupt	61
4.1.65	Parameter	EthCtrlEnableWakeUpInterrupt	61
4.1.66	Parameter	EthCtrlEnableTimerInterrupt	62
4.1.67	Reference	ETH_E_ERR	62
4.1.68	Container	EthCtrlConfigReceptionOptions	62
4.1.69	Parameter	ENET_RX_CONFIG_ENABLE_PAYLOAD_LEN_CHECK	63
4.1.70	Parameter	ENET_RX_CONFIG_STRIP_CRC_FIELD	63
4.1.71	Parameter	ENET_RX_CONFIG_FORWARD_PAUSE_FRAMES	64
4.1.72	Parameter	ENET_RX_CONFIG_REMOVE_PADDING	64
4.1.73	Parameter	ENET_RX_CONFIG_ENABLE_FLOW_CONTROL	64
4.1.74	Parameter	ENET_RX_CONFIG_REJECT_BROADCAST_FRAMES	65
4.1.75	Parameter	ENET_RX_CONFIG_ENABLE_PROMISCUOUS_MODE	65
4.1.76	Parameter	ENET_RX_CONFIG_ENABLE_MII_LOOPBACK	66
4.1.77	Container	EthCtrlConfigTransmissionOptions	66
4.1.78	Parameter	ENET_TX_CONFIG_DISABLE_CRC_APPEND	66
4.1.79	Parameter	ENET_TX_CONFIG_ENABLE_MAC_ADDR_INSERTION	67
4.1.80	Container	EthCtrlConfigRxAcceleratorOptions	67
4.1.81	Parameter	ENET_RX_ACCEL_REMOVE_PAD	67
4.1.82	Parameter	ENET_RX_ACCEL_ENABLE_IP_CHECK	68

4.1.83	Parameter	ENET_RX_ACCEL_ENABLE_PROTO_CHECK	68
4.1.84	Parameter	ENET_RX_ACCEL_ENABLE_MAC_CHECK	69
4.1.85	Parameter	ENET_RX_ACCEL_ENABLE_SHIFT16	69
4.1.86	Container	EthCtrlConfigTxAcceleratorOptions	69
4.1.87	Parameter	ENET_TX_ACCEL_ENABLE_SHIFT16	70
4.1.88	Parameter	ENET_TX_ACCEL_INSERT_IP_CHECKSUM	70
4.1.89	Parameter	ENET_TX_ACCEL_INSERT_PROTO_CHECKSUM	71
4.1.90	Container	EthCtrlConfigEgress	71
4.1.91	Reference	EthCtrlConfigEgressLastSchedulerRef	71
4.1.92	Container	EthCtrlConfigEgressFifo	72
4.1.93	Parameter	EthCtrlConfigEgressFifoBufLenByte	72
4.1.94	Parameter	EthCtrlConfigEgressFifoBufTotal	73
4.1.95	Parameter	EthCtrlConfigEgressFifoIdx	73
4.1.96	Parameter	EthCtrlConfigEgressFifoCallback	74
4.1.97	Parameter	EthCtrlConfigEgressFifoPriorityAssignment	74
4.1.98	Container	EthCtrlConfigScheduler	74
4.1.99	Container	EthCtrlConfigSchedulerPredecessor	75
4.1.100	Parameter	EthCtrlConfigSchedulerPredecessorOrder	75
4.1.101	Reference	EthCtrlConfigSchedulerPredecessorRef	76
4.1.102	Container	EthCtrlConfigShaper	76
4.1.103	Parameter	EthCtrlConfigShaperIdleSlope	76
4.1.104	Parameter	EthCtrlConfigShaperMaxCredit	77
4.1.105	Parameter	EthCtrlConfigShaperMinCredit	77
4.1.106	Reference	EthCtrlConfigShaperPredecessorFifoRef	78
4.1.107	Container	EthCtrlConfigShaperVendorSpecific	78
4.1.108	Parameter	EthCtrlConfigHiCredit	78
4.1.109	Parameter	EthCtrlConfigLoCredit	79
4.1.110	Container	EthCtrlConfigIngress	79
4.1.111	Container	EthCtrlConfigIngressFifo	80
4.1.112	Parameter	EthCtrlConfigIngressFifoBufLenByte	80
4.1.113	Parameter	EthCtrlConfigIngressFifoBufTotal	81
4.1.114	Parameter	EthCtrlConfigIngressFifoIdx	81
4.1.115	Parameter	EthCtrlConfigIngressFifoCallback	81
4.1.116	Parameter	EthCtrlConfigIngressFifoPriorityAssignment	82
4.1.117	Container	EthDemEventParameterRefs	82
4.1.118	Reference	ETH_E_ACCESS	83
4.1.119	Reference	ETH_E_RX_FRAMES_LOST	83
4.1.120	Reference	ETH_E_CRC	84
4.1.121	Reference	ETH_E_UNDERSIZEFRAME	84
4.1.122	Reference	ETH_E_OVERSIZEFRAME	85
4.1.123	Reference	ETH_E_ALIGNMENT	85
4.1.124	Reference	ETH_E_SINGLECOLLISION	85
4.1.125	Reference	ETH_E_MULTIPLECOLLISION	86
4.1.126	Reference	ETH_E_LATECOLLISION	86
4.1.127	Container	EthCtrlConfigSpiConfiguration	87
4.1.128	Parameter	EthCtrlConfigSpiCommTimeout	87
4.1.129	Parameter	EthCtrlConfigSpiChunkPayloadSize	88
4.1.130	Parameter	EthCtrlConfigSpiCommRetries	88
4.1.131	Parameter	EthCtrlConfigSpiEnableControlDataProtection	89
4.1.132	Parameter	EthCtrlConfigSpiEnableRxCSAlign	89
4.1.133	Parameter	EthCtrlConfigSpiEnableRxCutThrough	90
4.1.134	Parameter	EthCtrlConfigSpiEnableRxZeroAlign	90
4.1.135	Parameter	EthCtrlConfigSpiEnableTransmitDataHdrSequence	91
4.1.136	Parameter	EthCtrlConfigSpiEnableTxChecksum	91

4.1.137 Parameter EthCtrlConfigSpiEnableTxCutThrough	92
4.1.138 Parameter EthCtrlConfigSpiSelectTimeStamp	92
4.1.139 Parameter EthCtrlConfigSpiTransmitCreditThreshold	93
4.1.140 Container EthCtrlConfigSpiSequence	93
4.1.141 Parameter EthCtrlConfigSpiAccessSynchronous	94
4.1.142 Reference EthCtrlConfigSpiSequenceName	94
4.1.143 Container CommonPublishedInformation	95
4.1.144 Parameter ModuleId	95
4.1.145 Parameter VendorId	95
4.1.146 Parameter VendorApiInfix	96
4.1.147 Parameter ArReleaseMajorVersion	97
4.1.148 Parameter ArReleaseMinorVersion	97
4.1.149 Parameter ArReleaseRevisionVersion	97
4.1.150 Parameter SwMajorVersion	98
4.1.151 Parameter SwMinorVersion	98
4.1.152 Parameter SwPatchVersion	99
5 Module Index	100
5.1 Software Specification	100
6 Data Structure Index	101
6.1 Data Structures	101
7 Module Documentation	102
7.1 ETH Driver	102
7.1.0.1 Functions Expectation called for transmission in polling mode:	102
7.2 ENET Driver	103
7.2.1 Detailed Description	103
7.2.1.1 Platform-specific prerequisites	103
7.2.1.2 How to use the ENET driver in your application	103
7.2.1.3 How to enable zero-copy optimization	104
7.2.1.4 How to calculate size of ethernet buffer areas	104
7.2.1.5 How to set up MAC address filters	105
7.2.1.6 Examples:	105
7.2.2 Data Structure Documentation	111
7.2.2.1 struct Enet_Ip_BufferType	111
7.2.2.2 struct Enet_Ip_BufferDescriptorType	111
7.2.2.3 struct Enet_Ip_BufferConfigType	112
7.2.2.4 struct Enet_Ip_ConfigType	114
7.2.2.5 struct Enet_Ip_StateType	116
7.2.2.6 struct Enet_Ip_TimeStampType	120
7.2.2.7 struct Enet_Ip_RxInfoType	121
7.2.2.8 struct Enet_Ip_TxInfoType	123
7.2.2.9 struct Enet_Ip_TimerConfigType	123
7.2.2.10 struct Enet_Ip_TimerChannelConfigType	124
7.2.2.11 struct Enet_Ip_TxOptionsType	125
7.2.3 Types Reference	125
7.2.3.1 Enet_Ip_CallbackType	126
7.2.3.2 Enet_Ip_ChCallbackType	126
7.2.3.3 Enet_Ip_TimerCallbackType	126
7.2.4 Enum Reference	126
7.2.4.1 Enet_Ip_StatusType	126
7.2.4.2 Enet_Ip_EventType	127
7.2.4.3 Enet_Ip_TxSpecialConfigType	127

7.2.4.4	Enet_Ip_TxAcceleratorType	127
7.2.4.5	Enet_Ip_RxAcceleratorType	127
7.2.4.6	Enet_Ip_CounterType	127
7.2.4.7	Enet_Ip_MiiSpeedType	127
7.2.4.8	Enet_Ip_PowerStateType	128
7.2.4.9	Enet_Ip_SysTimeCorrOffsetType	128
7.2.4.10	Enet_Ip_TimerChannelModeType	128
7.2.4.11	Enet_Ip_ChecksumInsControlType	128
7.2.4.12	Enet_Ip_MmfrOpType	129
7.2.4.13	Enet_Ip_FifoThresholdType	129
7.2.5	Function Reference	130
7.2.5.1	Enet_Ip_Init()	130
7.2.5.2	Enet_Ip_GetPowerState()	130
7.2.5.3	Enet_Ip_Deinit()	131
7.2.5.4	Enet_Ip_EnableController()	131
7.2.5.5	Enet_Ip_RecoverFilter()	131
7.2.5.6	Enet_Ip_DisableController()	131
7.2.5.7	Enet_Ip_SetSpeed()	132
7.2.5.8	Enet_Ip_GetTxBuff()	132
7.2.5.9	Enet_Ip_SendFrame()	133
7.2.5.10	Enet_Ip_ReadFrame()	133
7.2.5.11	Enet_Ip_ProvideRxBuff()	134
7.2.5.12	Enet_Ip_IsFrameAvailable()	135
7.2.5.13	Enet_Ip_GetTransmitStatus()	135
7.2.5.14	Enet_Ip_GetCounter()	136
7.2.5.15	Enet_Ip_ConfigCounters()	136
7.2.5.16	Enet_Ip_EnableMDIO()	136
7.2.5.17	Enet_Ip_MDIORead()	137
7.2.5.18	Enet_Ip_MDIOWrite()	137
7.2.5.19	Enet_Ip_MDIOReadMMD()	138
7.2.5.20	Enet_Ip_MDIOWriteMMD()	138
7.2.5.21	Enet_Ip_SetMacAddr()	139
7.2.5.22	Enet_Ip_GetMacAddr()	139
7.2.5.23	Enet_Ip_SetBroadcastForwardAll()	139
7.2.5.24	Enet_Ip_SetMulticastForwardAll()	140
7.2.5.25	Enet_Ip_AddDstAddrToHashFilter()	140
7.2.5.26	Enet_Ip_RemoveDstAddrFromHashFilter()	140
7.2.5.27	Enet_Ip_SetSysTimeCorr()	140
7.2.5.28	Enet_Ip_TimerSetCorrection()	141
7.2.5.29	Enet_Ip_TimerStop()	141
7.2.5.30	Enet_Ip_TimerEnableChannel()	141
7.2.5.31	Enet_Ip_TimerSetCompare()	142
7.2.5.32	Enet_Ip_TimerStart()	142
7.2.5.33	Enet_Ip_TimerInit()	142
7.2.5.34	Enet_Ip_TimerSet()	143
7.2.5.35	Enet_Ip_TimerGet()	143
7.2.5.36	Enet_Ip_EnableTxStoreAndForward()	143
7.2.5.37	Enet_Ip_SetFIFOThreshold()	143
7.2.5.38	Enet_Ip_SetTxFIFOWatermark()	144
7.3	Ethernet Driver	145
7.3.1	Detailed Description	145
7.3.2	Macro Definition Documentation	145
7.3.2.1	ETH_43_ENET_E_COMMUNICATION	145
7.3.3	Function Reference	145

7.3.3.1 Eth_43_ENET_Init()	146
7.3.3.2 Eth_43_ENET_SetControllerMode()	146
7.3.3.3 Eth_43_ENET_GetControllerMode()	147
7.3.3.4 Eth_43_ENET_TxTimeAwareShaperInit()	147
7.3.3.5 Eth_43_ENET_GetPhysAddr()	147
7.3.3.6 Eth_43_ENET_SetPhysAddr()	148
7.3.3.7 Eth_43_ENET_ProvideTxBuffer()	148
7.3.3.8 Eth_43_ENET_Transmit()	149
7.3.3.9 Eth_43_ENET_Receive()	150
7.3.3.10 Eth_43_ENET_TxConfirmation()	150

8 Data Structure Documentation 151

8.1 Enet_Ip_ManagementInfo Struct Reference	151
8.1.1 Detailed Description	151

Chapter 1

Revision History

Revision	Date	Author	Description
1.0	07.03.2025	NXP RTD Team	S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductors' AUTOSAR Ethernet Driver for S32K1.

AUTOSAR Ethernet Driver configuration parameters description can be found in the Tressos Configuration Plugin section. Deviations from the specification are described in the [Deviations from Requirements](#) section.

AUTOSAR Ethernet driver requirements and APIs are described in the Ethernet Driver Software Specification Document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k148_lqfp100
- s32k148_mapbga100 / MWCT1016S_mapbga100
- s32k148_lqfp144
- s32k148_lqfp176

All of the above microcontroller devices are collectively named as S32K1. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".

- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
DEM	Diagnostic Event Manager
DET	Default Error Tracer
ETH	Ethernet
ETHIF	Ethernet Interface
ETHTRCV	Ethernet Transceiver
ETHSWT	Ethernet Switch
MCU	Micro controller Unit
MII	Media Independent Interface
N/A	Not Available
RMII	Reduced Media Independent Interface
RAM	Random Access Memory

- The term "Ethernet Controller" is related to the hardware module providing the Ethernet functionality.
- The term "Ethernet Driver" is related to the software handling the Ethernet Controller.
- The term "Application" is used for the software utilizing the Ethernet Driver.

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
DEM	Diagnostic Event Manager
DET	Default Error Tracer
ETH	Ethernet
ETHIF	Ethernet Interface
ETHTRCV	Ethernet Transceiver
ETHSWT	Ethernet Switch
MCU	Microcontroller Unit
MII	Media Independent Interface
N/A	Not Available
RMII	Reduced Media Independent Interface
RAM	Random Access Memory

- The term "Ethernet Controller" is related to the hardware module providing the Ethernet functionality.
- The term "Ethernet Driver" is related to the software handling the Ethernet Controller.
- The term "Application" is used for the software utilizing the Ethernet Driver.

2.5 Reference List

#	Title	Version
1	Specification of Driver	AUTOSAR Release R21-11

#	Title	Version
2	S32K1xx Reference Manual	Rev.14.1, 01/2024
3	S32M24x Reference Manual	Rev. 5, 12/2024
4	S32K1xx Data Sheet	Rev. 14, 08/2021
5	S32M2xx Data Sheet	Rev. 7, 12/2024
6	S32K116 Errata	Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024
7	S32K118 Errata	Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024
8	S32K142 Errata	Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024
9	S32K144 Errata	Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024
10	S32K144W Errata	Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024
11	S32K146 Errata	Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024
12	S32K148 Errata	Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024
13	S32M242 Errata	Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024
14	S32M244 Errata	Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)
- [Usage in non AUTOSAR context](#)

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR R21-11 Ethernet Driver Software Specification document (See Table [Reference List](#))

3.2 Driver Design Summary

The Ethernet Driver controls the Ethernet Media Access Controller (ENET) module of the S32K1 device. It provides the following features:

- Configuration and initialization of the Ethernet Controller
- Switching the Ethernet Controller on and off
- Reception and transmission of Ethernet frames
- Access to the Ethernet Controller's RMON counters (RxStats, TxStats and TxErrorCounterValues)
- Access to the Ethernet Transceiver device registers through MDIO
- Handling of the Ethernet Controller's interrupt requests
- Half and full duplex operation support
- Various PHY Interfaces (10/100 Mbps on MII and RMII)
- Time Synchronization over Ethernet (The gPTP stack has to be provided by the upper layers)
- Checksum hardware offloading for IPv4/IPv6 with TCP, UDP or ICMP

3.3 Hardware Resources

For the Ethernet driver, there are just S32K148 supported.

The hardware module configured by the Ethernet driver is ENET (Ethernet Media Access Controller). It has a single hardware instance (ENET_0).

The Ethernet controller to microcontroller pin mapping can be done using the file **S32K148_IO_Signal_↔ Description_Input_Multiplexing.xlsx** attached to the Reference Manual.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR ETH Driver software specification in some places. The table below identifies the AUTOSAR requirements that are not implemented or out of scope for the ETH Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently or out of scope for the ETH driver.

Requirement	Status	Description	Notes
SWS_Eth_00263	N/S	If the configuration parameter Eth↔CtrlConfigSwBufferHandling is set to TRUE, then the optional SW buffer handling shall be enabled.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00264	N/S	If the config parameter EthCtrl↔ConfigSwBufferHandling is set to T↔RUE, then each SW FIFO shall have the total amount of elements given by EthCtrlConfigEgressFifoBufTotal (). Each element shall be of type Eth↔_BufIdxType.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00265	N/S	"All SW FIFOs shall follow the criteria listed here: Each SW FIFO shall be filled and read out according to F↔IFO principles. The SW FIFOs shall support independent configuration regardless of any settings on the rest of SW FIFOs."	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress

Requirement	Status	Description	Notes
SWS_Eth_00266	N/S	SW FIFOs shall be iterated and their credits account be updated in the following way and order: - Credits are only accumulated for SW FIFOs which have at least one message queued inside them. Empty SW FIFOs do not accumulate credits and their credits counter shall be set to 0. - Iterate through all SW FIFOs, starting at the highest priority SW FIFO and descending, and add the amount of credits accumulated since the last Eth_↵MainFunction() call. The amount of credits accumulated is given by Eth_↵CtrlConfigShaperIdleSlope. - If a SW FIFO reaches EthCtrlConfigShaper_↵MaxCredit then the credit accumulation shall stop at that point and the next SW FIFO in the row is handled.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00267	N/S	If Eth_ProvideTxBuffer() is called and EthCtrlConfigSwBufferHandling is set to TRUE, a tuple of BuffIdx pointer to the SW buffer pool (which is returned) and priority (provided by argument of the current function call) shall be stored.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00268	N/S	When Eth_Transmit() is called and EthCtrlConfigSwBufferHandling is set to TRUE, the given BuffIdx pointer shall be assigned to the SW FIFO with the EthCtrlConfigEgressFifoPriority_↵Assignment which matches the priority given previously by the previous Eth_↵_ProvideTxBuffer() call (see SWS_↵Eth_00267).	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00269	N/S	Upon calling Eth_Transmit(), messages from the SW FIFOs shall be moved to the HW FIFO as described in SWS_Eth_00271.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00270	N/S	In the context of Eth_MainFunction(), the following actions shall be executed in the given order: - All SW FIFOs shall be iterated and their credits account updated as specified in SWS_↵Eth_00266. - All SW FIFOs shall be iterated and checked for messages which are ready for transmission. - For each SW FIFO iterated, transmission shall be attempted as specified in S_↵WS_Eth_00271.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress

Requirement	Status	Description	Notes
SWS_Eth_00271	N/S	Messages queued inside SW FIFOs shall be moved to the HW FIFO in the following way and order: - Loop through each SW FIFO, starting at the highest priority in descending order. - Move the first message inside a SW FIFO whose credit account is at least EthCtrlConfigShaperMinCredit to the HW FIFO. - If EthTrcvPhysLayerPLCAMaxBurstCount is set to 0 then only one message is moved to the HW FIFO and the iteration to the next SW FIFOs is stopped. - Reduce the SW FIFOs credits based on its EthCtrlConfigShaperSendSlope configuration. - If EthTrcvPhysLayerPLCAMaxBurstCount is higher than 0 then proceed on top as specified in SWS_Eth_00272.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00272	N/S	If EthTrcvPhysLayerPLCAMaxBurstCount is higher than 0, as many messages as EthTrcvPhysLayerPLCAMaxBurstCount indicates shall be moved additionally to the HW FIFO. The selection of each message shall be based on the requirements in SWS_Eth_00271.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00273	N/S	If Clause 45 registers need to be written via this access mechanism, the API shall use the register 13 and 14 to access them as explicitly specified by the annex 22D [20].	Propose to reject: We provide extension function for Clause45 support. Clause45 support through Clause22 operations can be done by the user.
SWS_Eth_00274	N/S	If Clause 45 registers need to be read via this access mechanism, the API shall use the register 13 and 14 to access them as explicitly specified by the annex 22D [20].	Propose to reject: We provide extension function for Clause45 support. Clause45 support through Clause22 operations can be done by the user.
ECUC_Eth_00071	N/S	Name: EthCtrlConfigSwBufferHandling Parent Container: EthCtrlConfig Description: Enables / Disables SW buffer management Multiplicity: 1 Type: EcucBooleanParamDef Default value: - Post-Build Variant Value: false Value Configuration Class: Pre-compile time: X: All Variants Link time: - Post-build time: - Scope / Dependency: scope: local	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00285	N/S	If an off-chip Ethernet controller is used (i.e. MACPHY), the Ethernet controller module shall use services of other MCAL drivers (e.g. SPI).	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
SWS_Eth_00299	N/S	If the configuration parameter <code>EthCtrlConfigSwBufferHandling</code> is set to TRUE, then one SW FIFO shall be available per configured <code>EthCtrlConfigEgressFifo</code> .	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00298	N/S	If the configuration parameter <code>EthCtrlConfigSwBufferHandling</code> is set to TRUE, then each SW FIFO shall handle frames according to the configured priorities given by <code>EthCtrlConfigEgressFifoPriorityAssignment</code> aggregated by the according <code>EthCtrlConfigEgressFifo</code> . If no <code>EthCtrlConfigEgressFifoPriorityAssignment</code> is configured, then any priority shall be handled by this SW FIFO.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00297	N/S	If the config parameter <code>EthCtrlConfigSwBufferHandling</code> is set to TRUE, then a SW buffer shall be provided with a size according to all configured <code>EthCtrlConfigEgressFifo</code> 's. The size of each <code>EthCtrlConfigEgressFifo</code> shall be calculated in bytes by considering the following formula: $\text{size of one EthCtrlConfigEgressFifo} = \text{EthCtrlConfigEgressFifoBufTotal} * \text{EthCtrlConfigEgressFifoBufLenByte}$.	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00300	N/S	If the config parameter <code>EthCtrlConfigSwBufferHandling</code> is set to TRUE, then all SW FIFOs and SW buffer pools shall be initialized with '0'	SW buffer management is not needed, as the hardware already has hardware buffers for both egress and ingress
SWS_Eth_00287	N/S	If <code>EthCtrlEnableSpiInterface</code> is TRUE, the function shall process the write request as described in the TC6 [25].	No off-chip Ethernet controllers are in scope of this driver
SWS_Eth_00290	N/S	If <code>EthCtrlEnableSpiInterface</code> is TRUE, the function shall process the read request as described in the TC6 [25].	No off-chip Ethernet controllers are in scope of this driver
SWS_Eth_00292	N/S	If development error detection is enabled: the function shall check that the service <code>Eth_Init</code> was previously called. If the check fails, the function shall raise the development error <code>ETH_E_UNINIT</code> .	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
SWS_Eth_00293	N/S	If development error detection is enabled: the function shall check the parameter CtrlIdx for being valid. If the check fails, the function shall raise the development error ETH_E_INVALID_CTRL_IDX.	No off-chip Ethernet controllers are in scope of this driver
SWS_Eth_00294	N/S	If development error detection is enabled: the function shall check the parameter TxStats for being valid. If the check fails, the function shall raise the development error ETH_E_PARM_RAM_POINTER.	No off-chip Ethernet controllers are in scope of this driver
SWS_Eth_00295	N/S	The function Eth_GetSpiStatus shall be pre compile time configurable On/Off by the configuration parameter: EthCtrlEnableSpiInterface.	No off-chip Ethernet controllers are in scope of this driver
SWS_Eth_91012	N/S	"Service Name: Eth_GetSpiStatus (draft) Syntax: Std_ReturnType Eth_GetSpiStatus (uint8 CtrlIdx, Eth_SpiStatusType* SpiStatusType) Service ID [hex]: 0x1E Sync/Async: Synchronous Reentrancy: Non Reentrant Parameters (in): CtrlIdx: Index of the controller within the context of the Ethernet Driver Parameters (inout): None Parameters (out): SpiStatusType: MACPHY status Return value: Std_ReturnType: E_OK: success, E_NOT_OK: Status could not be obtained Description: Returns the status defined by OA TC6 [26] to identify if an error can occurred at the SPI interface.: Tags: atp.Status=draft Available via: Eth.h"	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
SWS_Eth_91013	N/S	Name: Eth_SpiStatusType (draft) Kind: Structure Elements: Spi← StatusRegister Type: uint32 Comment: Bit mapped status defined by OA TC6 [26] to notify following information:: (Pos : description)← : 0x00: Transmit_Protocol_← Error,; 0x01: Transmit_Buffer_← Overflow_Error,; 0x02: Transmit_← _Buffer_Underflow_Error,; 0x03: Receive_Buffer_Overflow_Error,; 0x04: Loss_Framing_error,; 0x05: Header_Error,; 0x06: Reset_← Complete,; 0x07: PHY_Interrupt,← : 0x08: Transmit_Timestamp Capture_Available_A,; 0x09← : Transmit_Timestamp Capture← _Available_B,; 0x0A: Transmit_← Timestamp Capture_Available_C,; 0x0B: Transmit_Frame_Check_← Sequence_Error,; 0x0C: Control← _Data_Protection_Error,; 0x0D - 0xFF: Reserved. Sync Type: boolean Comment: Synchronization configura- tion as defined in the OA TC6 [26]. TRUE: MACPHY has been reset and is not configured. FALSE: MACPHY is configured. BufferStatusTxCredit Type: uint8 Comment: Contains the number of consecutive transmitted data chunks of Ethernet frame the SPI host can write without overflow- ing the MAC. BufferStatusRxCredit Type: uint8 Comment: Contains the number of additional received data chunks of Ethernet frame currently available for the SPI host to read. Description: Returns the Spi status, errors and configuration state.: Tags: atp.Status=draft Available via: Eth.h	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00073	N/S	"Name: EthCtrlEnableSpiInterface Parent Container: EthCtrlConfig Description: This optional parameter enables the processing of control data and Ethernet frames over the SPI interface specific for MACPHY device. The use of this parameter implies the respect of the SPI protocol described in TC6 [26]. Multiplicity: 0..1 Type: EcucBooleanParamDef Default value: false Post-Build Variant Multiplicity: false Post-Build Variant Value: true Value Configuration Class: Pre-compile time: X: All Variants Link time: -- Post-build time: -- Scope / Dependency: scope: local"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00074	N/S	"Container Name: EthCtrlConfig SpiConfiguration Parent Container: EthCtrlConfig Description: SPI Interface configuration of one Ethernet controller (MACPHY use). Configured only if EthCtrlEnableSpiInterface is set to TRUE.: Tags: atp.Status=draft Post-Build Variant Multiplicity: false Configuration Parameters"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00079	N/S	"Name: EthCtrlConfigSpiChunkPayloadSize Parent Container: EthCtrlConfigSpiConfiguration Description: Configures the size of the payload chunks which will be transferred over the SPI interface. Note: The chunk is the basic element for data transaction over the SPI which can be a section of an Ethernet frame or management command. The configured value has to be a multiple of 8.: Tags: atp.Status=draft Multiplicity: 1 Type: EcucIntegerParamDef Range: 8 .. 64 Default value: 64 Post-Build Variant Value: true Value Configuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE."	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00075	N/S	"Name: EthCtrlConfigSpiComm↵ Retries Parent Container: EthCtrl↵ ConfigSpiConfiguration Description: Indicates the maximum number of communication retries in case of a failed SPI communication (applies both to timed out communication and to errors/NACK in the response data). If configured value is '0', no retry is allowed (communication is expected to succeed at first try).: Tags: atp.↵ Status=draft Multiplicity: 1 Type: EcucIntegerParamDef Range: 0 .. 255 Default value: -- Post-Build Variant Value: true Value Configuration Class: Pre-compile time: X: VARI↵ ANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local↵ : dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE. This parameter exists only if at least one SPI Sequence is referenced."	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00076	N/S	"Name: EthCtrlConfigSpiComm↵ Timeout Parent Container: EthCtrl↵ ConfigSpiConfiguration Description: Indicates the maximum time allowed to the Ethernet controller for replying (either positively or negatively) to a SPI command. Timeout is configured in seconds. Timeout value of '0' means that no specific timeout is to be used by Ethernet controller and the communication is executed at the best of the SPI HW capacity.: Tags: atp.↵ Status=draft Multiplicity: 1 Type: EcucFloatParamDef Range: [0 .. 0.1] Default value: -- Post-Build Variant Value: true Multiplicity Configuration Class: Pre-compile time: -- Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-↵ POST-BUILD Value Configuration Class: Pre-compile time: X: VARI↵ ANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local↵ : dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE. This parameter exists only if at least one SPI Sequence is referenced."	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00081	N/S	"Name: EthCtrlConfigSpiEnable↵ ControlDataProtection Parent Container: EthCtrlConfigSpi↵ Configuration Description: Enables the control data protection. When set, all control data written to and read from the MACPHY will be transferred with its complement for detection of bit errors as defined in OA TC6 [26]. FALSE: Control data read/write protection is disabled (unprotected). TRUE: Control data read/write rotection is enabled (protected).: Tags: atp.Status=draft Multiplicity: 1 Type: EcucBooleanParamDef De- fault value: false Post-Build Variant Value: false Value Configuration Class: Pre-compile time: X: All Variants Link time: -- Post-build time: -- Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00085	N/S	"Name: EthCtrlConfigSpiEnableRx↵ CSAlign Parent Container: EthCtrl↵ ConfigSpiConfiguration Description↵ : Configures the CSn Align Receive frame. TRUE: all received Ethernet frames data shall start at the begin- ning of the first receive data chunk pay- load following CSn assertion FALS↵ E: received frames may begin within any receive data chunk of the trans- action when this bit is clear.: Tags↵ : atp.Status=draft Multiplicity: 1 Type: EcucBooleanParamDef Default value: false Post-Build Variant Value↵ : true Value Configuration Class↵ : Pre-compile time: X: VARIA↵ NT-PRE-COMPILE Link time: X↵ : VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUIL↵ D Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is config- ured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00082	N/S	"Name: EthCtrlConfigSpiEnable↵ RxCutThrough Parent Container↵ : EthCtrlConfigSpiConfiguration Description: When supported by the HW, enables the cut through mode of frame from the network to the SPI host.: Tags: atp.Status=draft Multiplicity: 0..1 Type: Ecuc↵ BooleanParamDef Default value: false Post-Build Variant Multiplicity: true Post-Build Variant Value: true Multiplicity Configuration Class: Pre-compile time: X: VARIAN↵ T-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-B↵ UILD Value Configuration Class: Pre-compile time: X: VARIAN↵ T-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local↵ : dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00084	N/S	"Name: EthCtrlConfigSpiEnable↵ RxZeroAlign Parent Container↵ : EthCtrlConfigSpiConfiguration Description: Configures the zero-align receive frame. TRUE: all received Ethernet frames data shall be aligned to start at the beginning of any receive data chunk payload. FALSE: Received frames may begin anywhere within the receive data chunk payload.: Tags: atp.Status=draft Multiplicity: 1 Type: EcucBooleanParamDef De- fault value: false Post-Build Variant Value: true Value Configuration Class: Pre-compile time: X: VARI↵ ANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local↵ : dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00080	N/S	"Name: EthCtrlConfigSpiEnable↵ TransmitDataHdrSequence Parent Container: EthCtrlConfigSpi↵ Configuration Description: When supported by the HW, enables the transmit data sequence monitor- ing. FALSE: transmit data header sequence bit monitoring disabled. TRUE: transmit data header sequence bit monitoring enabled.: Tags↵ : atp.Status=draft Multiplicity: 0..1 Type: EcucBooleanParamDef De- fault value: false Post-Build Variant Multiplicity: true Post-Build Variant Value: true Multiplicity Configura- tion Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-↵ POST-BUILD Value Configuration Class: Pre-compile time: X: VARI↵ ANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local↵ : dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00086	N/S	"Name: EthCtrlConfigSpiEnable↵ TxChecksum Parent Container↵ : EthCtrlConfigSpiConfiguration Description: Configures the CSn Align Receive frame. TRUE: all received Ethernet frames data shall start at the beginning of the first receive data chunk payload following CSn assertion FALSE: received frames may begin within any receive data chunk of the transaction when this bit is clear.: Tags: atp.Status=draft Multiplicity: 0..1 Type: EcucBooleanParamDef Default value: false Post-Build Vari- ant Multiplicity: true Post-Build Variant Value: true Multiplicity Con- figuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-↵ POST-BUILD Value Configuration Class: Pre-compile time: X: VARI↵ ANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local↵ : dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00089	N/S	"Name: EthCtrlConfigSpiEnable↵ TxCutThrough Parent Container↵ : EthCtrlConfigSpiConfiguration Description: When supported by the HW, enables the cut through mode of frame from SPI host to the network.: Tags: atp.Status=draft Multiplicity: 0..1 Type: EcucBooleanParamDef Default value: false Post-Build Vari- ant Multiplicity: false Post-Build Variant Value: false Multiplicity Configuration Class: Pre-compile time: X: All Variants Link time: -- Post-build time: -- Value Configu- ration Class: Pre-compile time: X: All Variants Link time: -- Post-build time: -- Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00087	N/S	"Name: EthCtrlConfigSpiSelectTimestamp Parent Container: EthCtrlConfigSpiConfiguration Description: When timestamp supported by the HW, selects size and format of the timestamps. FALSE: 32-bits timestamps TRUE: 64-bit timestamps Tags: atp.Status=draft Multiplicity: 0..1 Type: EcucBooleanParamDef Default value: false Post-Build Variant Multiplicity: true Post-Build Variant Value: true Multiplicity Configuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Value Configuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00083	N/S	"Name: EthCtrlConfigSpiTransmitCreditThreshold Parent Container: EthCtrlConfigSpiConfiguration Description: Configures the minimum of available transmit credit before the writing IRQn is asserted. As per OA TC6, this information is notified by the TXC field. 0 = 1 credit 1 = 4 credits 2 = 8 credits 3 = 16 credits Tags: atp.Status=draft Multiplicity: 1 Type: EcucIntegerParamDef Range: 0 .. 3 Default value: 0 Post-Build Variant Value: true Value Configuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE."	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00077	N/S	"Container Name: EthCtrlConfigSpiSequence Parent Container: EthCtrlConfigSpiConfiguration Description: Container gives Ethernet controller driver information about one SPI sequence. One SPI sequence used by Ethernet controller driver is in exclusive use for it. No other driver is allowed to access this sequence. Ethernet controller driver may use one sequence to access n Ethernet controller hardware chips of the same type or n sequences are used to access one single Ethernet controller hardware chip. If a Ethernet controller hardware has no SPI interface, there is no instance of this container.: Tags: atp.Status=draft Configuration Parameters"	No off-chip Ethernet controllers are in scope of this driver
ECUC_Eth_00078	N/S	"Name: EthCtrlConfigSpiAccess Synchronous Parent Container: EthCtrlConfigSpiSequence Description: This parameter is used to define whether the access to the Spi sequence is synchronous or asynchronous. true: SPI access is synchronous. false: SPI access is asynchronous.: Tags: atp.Status=draft Multiplicity: 0..1 Type: EcucBooleanParamDef Default value: false Post-Build Variant Multiplicity: true Post-Build Variant Value: true Multiplicity Configuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Value Configuration Class: Pre-compile time: X: VARIANT-PRE-COMPILE Link time: X: VARIANT-LINK-TIME Post-build time: X: VARIANT-POST-BUILD Scope / Dependency: scope: local: dependency: This parameter is valid, if EthCtrlEnableSpiInterface is configured and set to TRUE"	No off-chip Ethernet controllers are in scope of this driver

Requirement	Status	Description	Notes
ECUC_Eth_00088	N/S	"Name: EthCtrlConfigSpiSequence↵ Name Parent Container: EthCtrl↵ ConfigSpiSequence Description: Refer- ence to a Spi sequence configuration container.: Tags: atp.Status=draft Multiplicity: 0..* Type: Symbolic name reference to [SpiSequence] Post- Build Variant Multiplicity: false Post- Build Variant Value: false Multiplic- ity Configuration Class: Pre-compile time: X: All Variants Link time: - - Post-build time: -- Value Configu- ration Class: Pre-compile time: X↵ : All Variants Link time: -- Post-build time: -- Scope / Dependency: scope: local: dependency: SpiSequence"	No off-chip Ethernet controllers are in scope of this driver

3.5 Driver Limitations

The Ethernet Driver has the following limitations:

- The API functions *Eth_ReadMii* and *Eth_WriteMii* (and their counterparts *Eth_ReadMmd* and *Eth_Write↵Mmd*) are synchronous instead of asynchronous due to a variety of reasons further detailed in [this](#) AUTOSAR ticket.
- The length of a single received frame (including the 14-bytes Ethernet frame header and the 4-bytes FCS) must be less than or equal to *EthCtrlConfigIngressFifoBufLenByte*.

3.6 Driver usage and configuration tips

3.6.1 External data buffer usage for transmission and reception

Ethernet driver allows the usage of data buffers which are allocated externally in the application. This approach removes the data copy from the external buffers to the internal buffers which are by default used by the driver for transfers. In order to enable the External data buffers for Ethernet drive, the following configurations are necessary:

- Choose which data buffers are defined externally: data buffers for TX, RX or both and disable in configuration the following nodes:
 1. *EthConfigSet/EthCtrlConfig/*/EthCtrlVendorSpecific/EthCtrlConfigGeneral/ EthCtrlAllocateTx↵DataBuffers* ,
 2. *EthConfigSet/EthCtrlConfig/*/EthCtrlVendorSpecific/EthCtrlConfigGeneral/ EthCtrlAllocateRx↵DataBuffers* ,
 3. both respectively.
- Allocate the data buffers at application level. The application must ensure that the address is aligned to 64 bytes. *VAR_ALIGN*(address, size) macro from RTD package can be used.
- The data buffers must be placed in a memory section. There are two options: cacheable and no-cacheable.

1. **cache management enabled** EthGeneral/EthGeneralVendorSpecific/EthEnableCacheManagement node is checked, meaning that the driver ensures cache management for the data buffers. In this case, data buffers must be placed in cacheable memory section delimited by *ETH_START_SEC_VAR_CLEARED_UNSPECIFIED* and *ETH_STOP_SEC_VAR_CLEARED_UNSPECIFIED*.
2. **cache management disabled** EthGeneral/EthGeneralVendorSpecific/EthEnableCacheManagement node is not checked. In this case, data buffers must be placed in no cacheable memory section delimited by *ETH_START_SEC_VAR_CLEARED_UNSPECIFIED_NO_CACHEABLE* and *ETH_STOP_SEC_VAR_CLEARED_UNSPECIFIED_NO_CACHEABLE*.

3.6.1.1 Code snippets for External Buffers feature

The following code snippet illustrates how to allocate external data buffers in an application using both HLD Ethernet driver and IP Enet driver. The data buffers are also placed in a no-cacheable or cacheable memory section depending on the macro *ENET_HAS_CACHE_MANAGEMENT* which is generated based on the configuration mentioned above. *#if (ENET_HAS_CACHE_MANAGEMENT == STD_ON)*

```
#define ETH_START_SEC_VAR_CLEARED_UNSPECIFIED

#include "Eth_MemMap.h"

#else

#define ETH_START_SEC_VAR_CLEARED_UNSPECIFIED_NO_CACHEABLE

#include "Eth_MemMap.h"

#endif

VAR_ALIGN(uint8 RxFrameBuffer[ENET_0_MAX_RXBUFF_SUPPORTED *
ENET_0_MAX_RXBUFFLEN_SUPPORTED], FEATURE_ENET_BUFFDESCR_ALIGNMENT_BYTES)

VAR_ALIGN(uint8 FrameBuffer[ENET_0_MAX_TXBUFF_SUPPORTED *
ENET_0_MAX_TXBUFFLEN_SUPPORTED], FEATURE_ENET_BUFFDESCR_ALIGNMENT_BYTES)

#if (ENET_HAS_CACHE_MANAGEMENT == STD_ON)
#define ETH_STOP_SEC_VAR_CLEARED_UNSPECIFIED
#include "Eth_MemMap.h"
#else
#define ETH_STOP_SEC_VAR_CLEARED_UNSPECIFIED_NO_CACHEABLE
#include "Eth_MemMap.h"
#endif
```

The next code snippet presents how to use the HLD driver APIs with External Buffer feature. The following usecase has the loopback mode enabled and one frame is sent using polling mode from EthController is received back on the same controller. *int main()*

```
{
    int i;

    Std_ReturnType returnStatus;

    uint16 BufferLength;
```

```
Eth_RxStatusType Status;
```

```
/* 1. Call the init functions for the prerequisite modules and for eth driver*/
```

```
/* 2. Set the external RX buffers addresses inside module descriptors for all the rings .
```

This must be done always before enabling the controller, otherwise there could be frame loss and memory corruption due to missing data buffers on the reception./*

```
for (i=0; i < ENET_0_MAX_RXBUFF_SUPPORTED; i++)
```

```
{
```

```
Eth_ProvideRxBuffer(EthController, i, RxFrameBuffer + i*ENET_0_MAX_RXBUFFLEN_SUPPORTED);
```

```
}
```

```
/* 3. Enable the controller */
```

/ 4. For TX data buffers the complete frame must contain: ETHERNET HEADER (DST MAC + SRC MAC + FRAME TYPE) + DATA PAYLOAD:*

*The information must be copied in the buffer by the application before sending a frame. */*

```
/* 5. Call the Eth_SendFrame to send a buffer */
```

```
returnStatus = Eth_SendFrame(EthController, Ring, TxFrameBuffer, &BufferLength, TRUE);
```

```
/* 6. Wait frames to be transmitted. */
```

```
Eth_TxConfirmation(EthController);
```

```
/* 7. Wait for frames to be received back. */
```

```
do
```

```
{
```

```
Eth_Receive(EthController, 0U, &Status);
```

```
} while (Status == ETH_NOT_RECEIVED);
```

```
}
```

For the Enet IP driver, the same code flow must be used, but the functions used are the followings (in order):

1. void `Enet_Ip_ProvideRxBuff(uint8 Instance, uint8 Ring, const Enet_Ip_BufferType * Buff);`
2. `Enet_Ip_StatusType Enet_Ip_SendFrame(uint8 Instance, uint8 Ring, const Enet_Ip_BufferType * Buff, const Enet_Ip_TxOptionsType * Options);`
3. `Enet_Ip_StatusType Enet_Ip_GetTransmitStatus(uint8 Instance, uint8 Ring, const Enet_Ip_BufferType * Buff, Enet_Ip_TxInfoType * Info);`
4. `Enet_Ip_StatusType Enet_Ip_ReadFrame(uint8 Instance, uint8 Ring, Enet_Ip_BufferType * Buff, Enet_Ip_RxInfoType * Info);` and `void Enet_Ip_ProvideRxBuff(uint8 Instance, uint8 Ring, const Enet_Ip_BufferType * Buff);`. The last parameter of `Enet_Ip_ProvideRxBuff` must be the `Buff` parameter returned from `Enet_Ip_ReadFrame`.

Note

The detailed descriptions for the data types, driver functions can be found in the Module Documentation chapter of this manual.

The proper infix must be used for the HLD driver APIs, macros, data types and memory sections

3.7 Runtime errors

The Ethernet driver generates the following DEM extended production errors at runtime:

3.7.1 Standardized Extended Production Errors

Function	Error code	Condition triggering the error
Eth_[infix]_MainFunction	ETH_E_ACCESS	Controller access failed
Eth_[infix]_MainFunction	ETH_E_RX_FRAMES_LOST	Rx frame lost detected
Eth_[infix]_MainFunction	ETH_E_CRC	CRC failure detected
Eth_[infix]_MainFunction	ETH_E_UNDERSIZEFRAME	Undersized frame detected
Eth_[infix]_MainFunction	ETH_E_OVERSIZEFRAME	Oversized frame detected
Eth_[infix]_MainFunction	ETH_E_ALIGNMENT	Alignment error detected
Eth_[infix]_MainFunction	ETH_E_SINGLECOLLISION	Single collision detected
Eth_[infix]_MainFunction	ETH_E_MULTIPLECOLLISION	Multiple collision detected
Eth_[infix]_MainFunction	ETH_E_LATECOLLISION	Late collision detected

3.7.2 Vendor-Specific Extended Production Errors

Function	Error code	Condition triggering the error
Eth_Ipw_EventIrqCallback	ETH_E_ERR	Payload receive error, Collision retry limit reached, Late collision detected,

| AXI Bus Error detected, Babbling transmit error, Babbling receive error, Transmit FIFO underrun.

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like: *#define <Mip>Conf_<Container_ShortName>_<Container_ID>*

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing #ifdefs arguments).

3.9 Usage in non AUTOSAR context

For the non-ASR environments, both High-Level and IP layer interfaces are exposed and can be used by software units above. These may consist of either higher level NXP SW deliverables, like stacks and middleware, or the application code itself.

The goal of the product is to be versatile, easy to use and integrate with other SW environments, offering both standardized and optimized low-level interfaces.

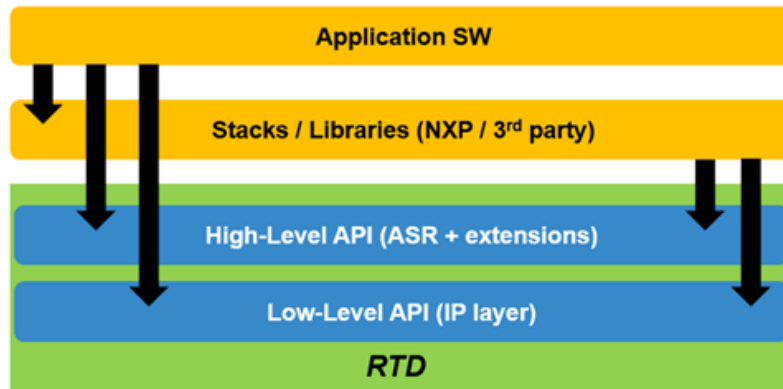


Figure 3.1 RTD Usage in non-ASR Context

In the non-AUTOSAR environment, there are no specific ECU states defined, but an external assumption for the application is that the correct sequence of initialization takes place prior to calling any runtime APIs.

The constraints considered for defining the APIs in the non-AUTOSAR environment are:

- Hardware resources used by the drivers must be properly initialized prior to runtime usage by calling the initialization function with configuration parameters generated by the configuration tools
- All dependencies must be initialized prior to the usage of a driver that depends on them; this translates to a general initialization sequence applicable for all applications, as follows:
 1. Initialization of the device clock tree so other periphery can be initialized
 2. Configuration of the pinout so peripherals with external connections can communicate with the outside world
 3. Configuration of the common resources (e.g. DMA channels, memory regions, access rights, etc) prior to the usage of any runtime functionality that has to benefit from this overall SoC configuration
 4. Initialization of specific drivers
 5. Runtime functionality
 6. Deinitialization (inverse initialization order)

High-level and IP layer functionalities cannot be used simultaneously over the same hardware resource. The granularity of the resource must be defined specifically for each driver, but as a general rule the default granularity considered is the peripheral instance (e.g. Gpt high level APIs cannot be called interchangeably with Pit_Ip low level APIs over the same PIT hardware instance).

The diagram below shows a generic, high-level sequence of calls to RTD interfaces in the non-AUTOSAR scenario:

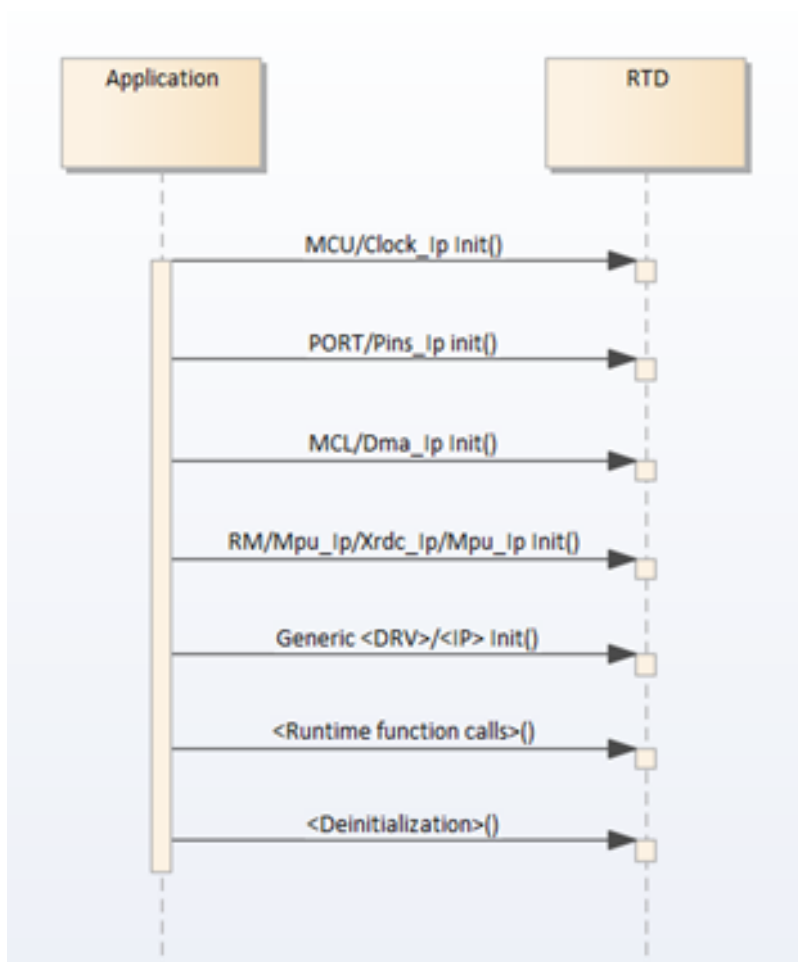


Figure 3.2 Non-AUTOSAR Sequence diagram

Chapter 4

Configuration Parameters

RTD provides configuration schema for the high level driver in EB Tresos proprietary format (**.xdm**), S32CT proprietary format (**.component**), but also in a standardized AUTOSAR format (**.epd**). For the low level driver configuration only the S32CT proprietary format (**.component**) is supported. The configuration data of the project can be saved in EB Tresos proprietary format (**.xdm**) and AUTOSAR standardized format (**.epc**) from EB Tresos and in S32CT proprietary format (**.mex**) and AUTOSAR standardized format (**.epc**) from S32 Design Studio. RTD provides templates for parsing the configuration data and generating configuration structures from it in XPath format, to be used with the EB Tresos generator and in JS format, to be used with the S32CT generator. Since the generation process is not standardized by AUTOSAR, only the two mentioned tools can be used with the provided templates. Using the same input configuration data, the same configuration structures will be obtained using any of the generators.

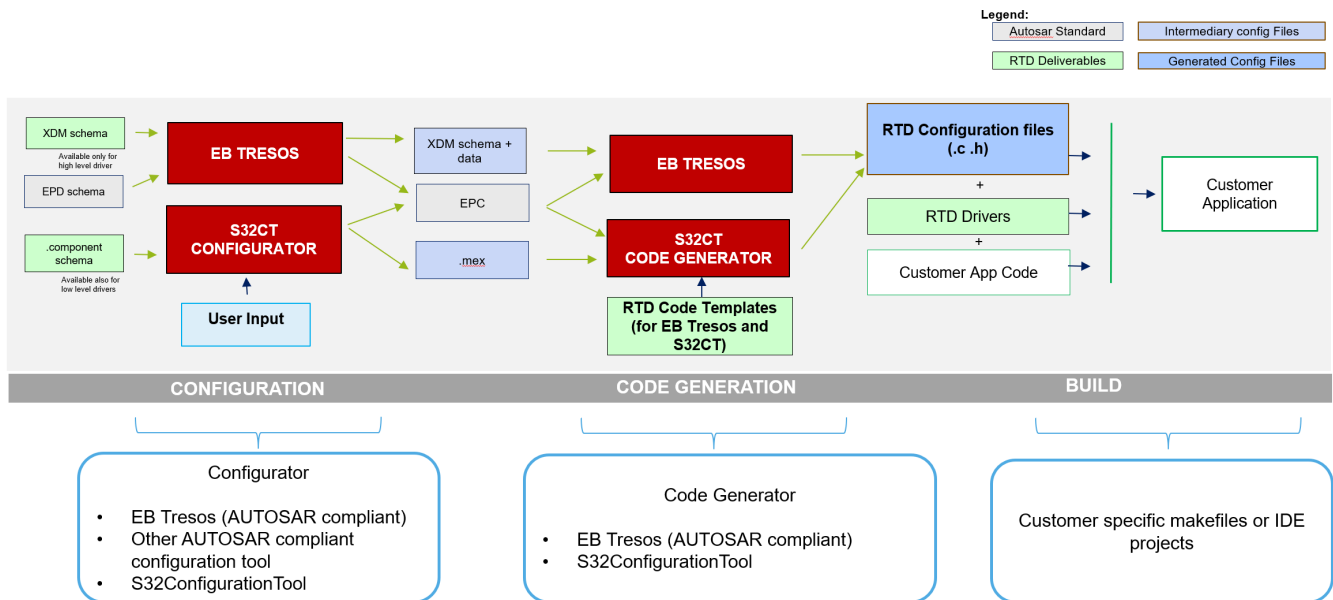


Figure 4.1 Configuration generation flow

- [Eth](#)

4.1 Eth

- Module [Eth](#)

- Container [EthGeneral](#)
 - * Parameter [EthDevErrorDetect](#)
 - * Parameter [EthGetDropCountApi](#)
 - * Parameter [EthGetEtherStatsApi](#)
 - * Parameter [EthGetCounterValuesApi](#)
 - * Parameter [EthGetRxStatsApi](#)
 - * Parameter [EthGetTxStatsApi](#)
 - * Parameter [EthGetTxErrorCounterValuesApi](#)
 - * Parameter [EthSendMultiBufferFrameApi](#)
 - * Parameter [EthGlobalTimeSupport](#)
 - * Parameter [EthVersionInfoApi](#)
 - * Parameter [EthIndex](#)
 - * Parameter [EthMainFunctionPeriod](#)
 - * Parameter [EthMaxCtrlsSupported](#)
 - * Parameter [EthMaxBuffersInMultiBufferFrame](#)
 - * Reference [EthEcucPartitionRef](#)
 - * Container [EthCtrlOffloading](#)
 - Parameter [EthCtrlEnableOffloadChecksumICMP](#)
 - Parameter [EthCtrlEnableOffloadChecksumIPv4](#)
 - Parameter [EthCtrlEnableOffloadChecksumTCP](#)
 - Parameter [EthCtrlEnableOffloadChecksumUDP](#)
 - * Container [EthGeneralVendorSpecific](#)
 - Parameter [EthDisableDemEventDetect](#)
 - Parameter [EthEnableUserModeSupport](#)
 - Parameter [EthMultiPartitionSupport](#)
 - Parameter [EthUpdatePhysAddrFilterApi](#)
 - Parameter [EthSwtManagementSupportApi](#)
 - Parameter [EthCoalescingInterrupt](#)
 - Parameter [EthTimeoutMethod](#)
 - Parameter [EthTimeoutDuration](#)
 - Parameter [EthEnableCacheManagement](#)
- Container [EthConfigSet](#)
 - * Container [EthCtrlConfig](#)
 - Parameter [EthCtrlEnableMii](#)
 - Parameter [EthCtrlEnableMmd](#)
 - Parameter [EthCtrlEnableRxInterrupt](#)

- Parameter [EthCtrlEnableTxInterrupt](#)
- Parameter [EthCtrlIdx](#)
- Parameter [EthCtrlMacLayerType](#)
- Parameter [EthCtrlMacLayerSubType](#)
- Parameter [EthCtrlMacLayerSpeed](#)
- Parameter [EthCtrlPhyAddress](#)
- Parameter [EthCtrlEnableSpiInterface](#)
- Parameter [EthCtrlConfigSwBufferHandling](#)
- Reference [EthCtrlEcucPartitionRef](#)
- Container [EthCtrlVendorSpecific](#)
 - Container [EthCtrlConfigGeneral](#)
 - Parameter [EthCtrlEnableAtInit](#)
 - Parameter [EthCtrlAllocateTxDataBuffers](#)
 - Parameter [EthCtrlAllocateRxDataBuffers](#)
 - Parameter [EthCtrlEthIfIdx](#)
 - Parameter [EthEthTrcvDriverVendorIdAndApiInfix](#)
 - Parameter [EthEthSwtDriverVendorIdAndApiInfix](#)
 - Parameter [EthDuplexMode](#)
 - Parameter [EthMaxFrameLength](#)
 - Parameter [EthTimeStampRequiredAccuracy](#)
 - Parameter [EthTimeStampCorrectionCounter](#)
 - Reference [EthModuleReferenceClock](#)
 - Reference [EthTimeStampReferenceClock](#)
 - Container [EthCtrlConfigInterrupts](#)
 - Parameter [EthCtrlConfigEventCallback](#)
 - Parameter [EthCtrlConfigTimeStampCallback](#)
 - Parameter [EthCtrlEnableErrorInterrupt](#)
 - Parameter [EthCtrlEnableWakeUpInterrupt](#)
 - Parameter [EthCtrlEnableTimerInterrupt](#)
 - Reference [ETH_E_ERR](#)
 - Container [EthCtrlConfigReceptionOptions](#)
 - Parameter [ENET_RX_CONFIG_ENABLE_PAYLOAD_LEN_CHECK](#)
 - Parameter [ENET_RX_CONFIG_STRIP_CRC_FIELD](#)
 - Parameter [ENET_RX_CONFIG_FORWARD_PAUSE_FRAMES](#)
 - Parameter [ENET_RX_CONFIG_REMOVE_PADDING](#)
 - Parameter [ENET_RX_CONFIG_ENABLE_FLOW_CONTROL](#)

- Parameter [ENET_RX_CONFIG_REJECT_BROADCAST_FRAMES](#)
- Parameter [ENET_RX_CONFIG_ENABLE_PROMISCUOUS_MODE](#)
- Parameter [ENET_RX_CONFIG_ENABLE_MII_LOOPBACK](#)
- Container [EthCtrlConfigTransmissionOptions](#)
 - Parameter [ENET_TX_CONFIG_DISABLE_CRC_APPEND](#)
 - Parameter [ENET_TX_CONFIG_ENABLE_MAC_ADDR_INSERTION](#)
- Container [EthCtrlConfigRxAcceleratorOptions](#)
 - Parameter [ENET_RX_ACCEL_REMOVE_PAD](#)
 - Parameter [ENET_RX_ACCEL_ENABLE_IP_CHECK](#)
 - Parameter [ENET_RX_ACCEL_ENABLE_PROTO_CHECK](#)
 - Parameter [ENET_RX_ACCEL_ENABLE_MAC_CHECK](#)
 - Parameter [ENET_RX_ACCEL_ENABLE_SHIFT16](#)
- Container [EthCtrlConfigTxAcceleratorOptions](#)
 - Parameter [ENET_TX_ACCEL_ENABLE_SHIFT16](#)
 - Parameter [ENET_TX_ACCEL_INSERT_IP_CHECKSUM](#)
 - Parameter [ENET_TX_ACCEL_INSERT_PROTO_CHECKSUM](#)
- Container [EthCtrlConfigEgress](#)
 - Reference [EthCtrlConfigEgressLastSchedulerRef](#)
 - Container [EthCtrlConfigEgressFifo](#)
 - Parameter [EthCtrlConfigEgressFifoBufLenByte](#)
 - Parameter [EthCtrlConfigEgressFifoBufTotal](#)
 - Parameter [EthCtrlConfigEgressFifoIdx](#)
 - Parameter [EthCtrlConfigEgressFifoCallback](#)
 - Parameter [EthCtrlConfigEgressFifoPriorityAssignment](#)
 - Container [EthCtrlConfigScheduler](#)
 - Container [EthCtrlConfigSchedulerPredecessor](#)
 - Parameter [EthCtrlConfigSchedulerPredecessorOrder](#)
 - Reference [EthCtrlConfigSchedulerPredecessorRef](#)
 - Container [EthCtrlConfigShaper](#)
 - Parameter [EthCtrlConfigShaperIdleSlope](#)
 - Parameter [EthCtrlConfigShaperMaxCredit](#)
 - Parameter [EthCtrlConfigShaperMinCredit](#)
 - Reference [EthCtrlConfigShaperPredecessorFifoRef](#)
 - Container [EthCtrlConfigShaperVendorSpecific](#)
 - Parameter [EthCtrlConfigHiCredit](#)
 - Parameter [EthCtrlConfigLoCredit](#)

- Container [EthCtrlConfigIngress](#)
 - Container [EthCtrlConfigIngressFifo](#)
 - Parameter [EthCtrlConfigIngressFifoBufLenByte](#)
 - Parameter [EthCtrlConfigIngressFifoBufTotal](#)
 - Parameter [EthCtrlConfigIngressFifoIdx](#)
 - Parameter [EthCtrlConfigIngressFifoCallback](#)
 - Parameter [EthCtrlConfigIngressFifoPriorityAssignment](#)
- Container [EthDemEventParameterRefs](#)
 - Reference [ETH_E_ACCESS](#)
 - Reference [ETH_E_RX_FRAMES_LOST](#)
 - Reference [ETH_E_CRC](#)
 - Reference [ETH_E_UNDERSIZEFRAME](#)
 - Reference [ETH_E_OVERSIZEFRAME](#)
 - Reference [ETH_E_ALIGNMENT](#)
 - Reference [ETH_E_SINGLECOLLISION](#)
 - Reference [ETH_E_MULTIPLECOLLISION](#)
 - Reference [ETH_E_LATECOLLISION](#)
- Container [EthCtrlConfigSpiConfiguration](#)
 - Parameter [EthCtrlConfigSpiCommTimeout](#)
 - Parameter [EthCtrlConfigSpiChunkPayloadSize](#)
 - Parameter [EthCtrlConfigSpiCommRetries](#)
 - Parameter [EthCtrlConfigSpiEnableControlDataProtection](#)
 - Parameter [EthCtrlConfigSpiEnableRxCSAlign](#)
 - Parameter [EthCtrlConfigSpiEnableRxCutThrough](#)
 - Parameter [EthCtrlConfigSpiEnableRxZeroAlign](#)
 - Parameter [EthCtrlConfigSpiEnableTransmitDataHdrSequence](#)
 - Parameter [EthCtrlConfigSpiEnableTxChecksum](#)
 - Parameter [EthCtrlConfigSpiEnableTxCutThrough](#)
 - Parameter [EthCtrlConfigSpiSelectTimeStamp](#)
 - Parameter [EthCtrlConfigSpiTransmitCreditThreshold](#)
 - Container [EthCtrlConfigSpiSequence](#)
 - Parameter [EthCtrlConfigSpiAccessSynchronous](#)
 - Reference [EthCtrlConfigSpiSequenceName](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ModuleId](#)
 - * Parameter [VendorId](#)

Configuration Parameters

- * Parameter [VendorApiInfix](#)
- * Parameter [ArReleaseMajorVersion](#)
- * Parameter [ArReleaseMinorVersion](#)
- * Parameter [ArReleaseRevisionVersion](#)
- * Parameter [SwMajorVersion](#)
- * Parameter [SwMinorVersion](#)
- * Parameter [SwPatchVersion](#)

4.1.1 Module Eth

Configuration of the Eth (Ethernet Driver) module.

Included containers:

- [EthGeneral](#)
- [EthConfigSet](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-PRE-COMPILE, VARIANT-LINK-TIME, VARIANT-POST-BUILD

4.1.2 Container EthGeneral

General configuration of Ethernet Driver module.

Included subcontainers:

- [EthCtrlOffloading](#)
- [EthGeneralVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.3 Parameter EthDevErrorDetect

Enables / Disables development error detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.4 Parameter EthGetDropCountApi

Obsolete and kept only for backwards compatibility. The Eth_GetDropCount API has been replaced with Eth_GetCounterValues since R4.3.1. Use parameter EthGetCounterValuesApi instead.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.5 Parameter EthGetEtherStatsApi

Obsolete and kept only for backwards compatibility. The Eth_GetEtherStats API has been replaced with the Eth_GetTxStats and Eth_GetRxStats APIs since R4.3.1.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.6 Parameter EthGetCounterValuesApi

Enables / Disables Eth_GetCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.7 Parameter EthGetRxStatsApi

Enables / Disables Eth_GetRxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.8 Parameter EthGetTxStatsApi

Enables / Disables Eth_GetTxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.9 Parameter EthGetTxErrorCounterValuesApi

Enables / Disables Eth_GetTxErrorCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.10 Parameter EthSendMultiBufferFrameApi

Enables / Disables Eth_SendMultiBufferFrame API. Important: This feature will only be available for controllers configured to use external buffers.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.11 Parameter EthGlobalTimeSupport

Enables/Disables the GlobalTime APIs used amongst others by Global Time Synchronization over Ethernet.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.12 Parameter EthVersionInfoApi

Enables / Disables Eth_GetVersionInfo API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.1.13 Parameter EthIndex

Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	255
min	0

4.1.14 Parameter EthMainFunctionPeriod

The period between successive calls to the main function in seconds.

Ethernet driver does not require this information but the BSW scheduler.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0.001
max	65.535
min	0.001

4.1.15 Parameter EthMaxCtrlsSupported

Limits the total number of supported controllers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	1
min	1

4.1.16 Parameter EthMaxBuffersInMultiBufferFrame

The maximum number of buffers that can be used in a multi buffer transmission. Note: This field is only used in the context of *Eth_SendMultiBufferFrame* API.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16
max	255
min	0

4.1.17 Reference EthEcucPartitionRef

Maps the Ethernet driver to zero or multiple ECUC partitions to make the module's API available in this partition.

The Ethernet driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite

Property	Value
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.1.18 Container EthCtrlOffloading

Configuration of hardware offloading features.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.19 Parameter EthCtrlEnableOffloadChecksumICMP

Enables / Disables hardware offloading for ICMP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.20 Parameter EthCtrlEnableOffloadChecksumIPv4

Enables / Disables hardware offloading for IPv4 checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.21 Parameter EthCtrlEnableOffloadChecksumTCP

Enables / Disables hardware offloading for TCP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.22 Parameter EthCtrlEnableOffloadChecksumUDP

Enables / Disables hardware offloading for UDP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.23 Container EthGeneralVendorSpecific

General vendor-specific configuration of Ethernet Driver.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.24 Parameter EthDisableDemEventDetect

When this option is enabled, no DEM events are reported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.25 Parameter EthEnableUserModeSupport

When this parameter is enabled, the Eth module will adapt to run from User Mode, with the following measures:

(if applicable) a) configuring REG_PROT for the Eth Controllers so that the registers under protection eth be accessed from user

Configuration Parameters

mode by setting UAA bit in REG_PROT_GCR to 1.

(if applicable) b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

(if applicable) c) other module specific measures for more information, please see chapter 5.7 User Mode Support in IM

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.26 Parameter EthMultiPartitionSupport

Switches multi-partition support on or off:

False: For all variants, no EcucPartition shall be referenced in EthEcucPartitionRef.

True: For all variants, at least one EcucPartition needs to be referenced in EthEcucPartitionRef.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.27 Parameter EthUpdatePhysAddrFilterApi

Enables/Disables Eth_UpdatePhysAddrFilter API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.28 Parameter EthSwtManagementSupportApi

Enables / Disables the Switch Management APIs.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.29 Parameter EthCoalescingInterrupt

Enables / Disables Coalescing Interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Configuration Parameters

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.30 Parameter EthTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If SystemTimer or CustomTimer are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM', 'OSIF_COUNTER_DUMMY']

4.1.31 Parameter EthTimeoutDuration

The unit of measurement is given in number of microseconds.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295

Property	Value
min	1

4.1.32 Parameter EthEnableCacheManagement

Enable/disable support for cache management inside the driver.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.33 Container EthConfigSet

This container contains the configuration parameters and sub containers of the AUTOSAR Eth module.

Included subcontainers:

- [EthCtrlConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.34 Container EthCtrlConfig

Configuration of the individual controller

Included subcontainers:

- [EthCtrlVendorSpecific](#)
- [EthCtrlConfigEgress](#)
- [EthCtrlConfigIngress](#)
- [EthDemEventParameterRefs](#)
- [EthCtrlConfigSpiConfiguration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.35 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.1.36 Parameter EthCtrlEnableMmd

Enables / Disables Clause 45 Media Independent Interface (MII) for transceiver access.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.1.37 Parameter EthCtrlEnableRxInterrupt

Enables / Disables receive interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.1.38 Parameter EthCtrlEnableTxInterrupt

Enables / Disables transmit interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.1.39 Parameter EthCtrlIdx

Specifies the instance ID of the configured controller.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	0
min	0

4.1.40 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGMII', 'ETH_MAC_LAYER_TYPE_XXGMII']

4.1.41 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of the ethernet controller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	STANDARD
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.1.42 Parameter EthCtrlMacLayerSpeed

Defines the baud rate of the MAC layer.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_10M
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_10G']

4.1.43 Parameter EthCtrlPhyAddress

Specifies the unique 48-bit physical address (MAC address) of the controller in network byte order.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	66:55:44:33:22:11

4.1.44 Parameter EthCtrlEnableSpiInterface

This optional parameter enables the processing of control data and Ethernet frames over the SPI interface specific for MACPHY device. The use of this parameter implies the respect of the SPI protocol described in TC6 [26].

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.45 Parameter EthCtrlConfigSwBufferHandling

Enables / Disables SW buffer management

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.46 Reference EthCtrlEcucPartitionRef

Maps the Ethernet controller to zero or one ECUC partition.

The ECUC partition referenced is a subset of the ECUC partitions where the Ethernet driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.1.47 Container EthCtrlVendorSpecific

Vendor specific configuration of the individual controller

Included subcontainers:

- [EthCtrlConfigGeneral](#)
- [EthCtrlConfigInterrupts](#)
- [EthCtrlConfigReceptionOptions](#)
- [EthCtrlConfigTransmissionOptions](#)
- [EthCtrlConfigRxAcceleratorOptions](#)
- [EthCtrlConfigTxAcceleratorOptions](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.48 Container EthCtrlConfigGeneral

General Controller Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.49 Parameter EthCtrlEnableAtInit

Enabled - The controller is enabled at initialization.

Disabled - The controller is left disabled at initialization and must be later enabled by calling `Enet_Ip_EnableController`

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.50 Parameter EthCtrlAllocateTxDataBuffers

Enabled - The driver will allocate both Tx Buffer Descriptors and Tx Data Buffers.

Disabled - The driver will only allocate Tx Buffer Descriptors. The Tx Data Buffers will be allocated by the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.51 Parameter EthCtrlAllocateRxDataBuffers

Enabled - The driver will allocate both Rx Buffer Descriptors and Rx Data Buffers.

Disabled - The driver will only allocate Rx Buffer Descriptors. The Rx Data Buffers will be allocated by the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.1.52 Parameter EthCtrlEthIfIdx

Vendor specific: Defines the index of the controller in context of EthIf driver. This manual assignemnt of this value is only temporary workaround before proper solution is implemented.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.1.53 Parameter EthEthTrcvDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthTrcv driver, if any.

This is used to include the correct header file and to point to the EthTrcv callbacks.

Example: 43_PHYGROUPB for a 43_PHYGROUPB driver or leave it blank to include EthTrcv.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	

4.1.54 Parameter EthEthSwtDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthSwt driver, if any.

This is used to include the correct header file and to point to the EthSwt APIs.

Example: 43_SJA1110 for a 43_SJA1110 driver or leave it blank to include EthSwt.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	

4.1.55 Parameter EthDuplexMode

Defines the controller's duplex mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.1.56 Parameter EthMaxFrameLength

Maximum frame length starting at DA and includes the CRC at the end of the frame.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1518
max	1518
min	64

4.1.57 Parameter EthTimeStampRequiredAccuracy

Value given in nanoseconds.

Represents the minimum required accuracy for this gPTP device.

To ensure there's enough headroom for clock drifts from both the GM and Slave, a maximum drift percentage per host must be assumed.

That is, $TotalDriftPercentage = MaximumGmDriftPercentage + MaximumSlaveDriftPercentage$. Then the following inequality must be true:

$EthTimeStampReferenceClock * (1 + TotalDriftPercentage / 100) > (1 / EthTimeStampRequiredAccuracy) * 10^9$.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	40
max	4294967295
min	1

4.1.58 Parameter EthTimeStampCorrectionCounter

Value given in clock cycles.

Defines after how many timer clock cycles the correction counter should be reset and trigger a correction increment on the timer.

A value of 0 disables the correction counter and no corrections occur.

If remote frequency > local frequency, local timer need to slow down. So Correction increment must be smaller normal increment.

If remote frequency < local frequency, local timer need to speed up. So Correction increment must be bigger normal increment.

Calculation expression for Correction increment: $(\text{Origin} - (\text{EthTimeStampCorrectionCounter} * \text{normal Increment})) / ((\text{Local} / \text{normal increment}) - \text{EthTimeStampCorrectionCounter})$

With Origin and Local get value from type Eth_RateRatioType. Normal increment = $(1 / \text{local frequency}) * 10^9$.

Note: + Need to select value of EthTimeStampCorrectionCounter such that :

- Value Correction Increment is interger after calculation.
- This value also need to configure smaller than $(\text{Local} / \text{normal increment})$.
- Satisfied $\text{Origin} > \text{EthTimeStampCorrectionCounter} * \text{increment}$.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	1073741823
min	0

4.1.59 Reference EthModuleReferenceClock

Reference to the module clock for the ENET.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.1.60 Reference EthTimeStampReferenceClock

Reference to the Timestamp source clock for the ENET.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False

Property	Value
destination	/AUTOSAR/EcuDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.1.61 Container EthCtrlConfigInterrupts

Interrupt sources Implements

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.62 Parameter EthCtrlConfigEventCallback

Callback function invoked when one of the events: wakeup, parse, error is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Eth_Ipw_EventIrqCallback

4.1.63 Parameter EthCtrlConfigTimeStampCallback

Callback to be invoked when a timer event occurs

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Eth_Ipw_TimeStampCallback

4.1.64 Parameter EthCtrlEnableErrorInterrupt

Enable/disable error interrupts

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.1.65 Parameter EthCtrlEnableWakeUpInterrupt

Enable/disable wakeup interrupts

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.66 Parameter EthCtrlEnableTimerInterrupt

Enable/disable interrupts related to the timer

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.67 Reference ETH_E_ERR

Reference to the DemEventParameter which shall be issued when any payload receive/ collision retry limit reached/ late collision/ AXI Bus Error/ Babbling transmit/ Babbling receive/ Transmit FIFO underrun errors has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.68 Container EthCtrlConfigReceptionOptions

Reception special options Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.69 Parameter ENET_RX_CONFIG_ENABLE_PAYLOAD_LEN_CHECK

The core checks the frame's payload length with the frame length/type field. Errors are indicated in the EIR[PLC] field

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.70 Parameter ENET_RX_CONFIG_STRIP_CRC_FIELD

Specifies whether the CRC field of received frames is transmitted or stripped. NOTE: If padding function is enabled (PADEN = 1), CRCFWD is ignored and the CRC field is checked and always terminated and removed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.1.71 Parameter ENET_RX_CONFIG_FORWARD_PAUSE_FRAMES

If a TCP/IP, UDP/IP, or ICMP/IP frame is received that has a wrong TCP, UDP, or ICMP checksum, the frame is discarded. Discarding is only available when the RX FIFO operates in store and forward mode (RSFL cleared).

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.72 Parameter ENET_RX_CONFIG_REMOVE_PADDING

Padding is removed from received frames.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.73 Parameter ENET_RX_CONFIG_ENABLE_FLOW_CONTROL

If set, the receiver detects PAUSE frames. Upon PAUSE frame detection, the transmitter stops transmitting data frames for a given duration.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.74 Parameter ENET_RX_CONFIG_REJECT_BROADCAST_FRAMES

If set, frames with destination address (DA) equal to 0xFFFF_FFFF_FFFF are rejected unless the PROM field is set. If BC_REJ and PROM are set, frames with broadcast DA are accepted and the MISS (M) is set in the receive buffer descriptor.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.75 Parameter ENET_RX_CONFIG_ENABLE_PROMISCUOUS_MODE

All frames are accepted regardless of address matching.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.76 Parameter ENET_RX_CONFIG_ENABLE_MII_LOOPBACK

Transmitted frames are looped back internal to the device and transmit MII output signals are not asserted.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.1.77 Container EthCtrlConfigTransmissionOptions

Transmission special options Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.78 Parameter ENET_TX_CONFIG_DISABLE_CRC_APPEND

The transmitter does not append any CRC to transmitted frames, as it is expecting a frame with CRC from the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.79 Parameter ENET_TX_CONFIG_ENABLE_MAC_ADDR_INSERTION

The MAC overwrites the source MAC address with the programmed MAC address according to ADDSEL.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.80 Container EthCtrlConfigRxAcceleratorOptions

Enable/Disable available reception acceleration options.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.81 Parameter ENET_RX_ACCEL_REMOVE_PAD

Any bytes following the IP payload section of the frame are removed from the frame.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.82 Parameter ENET_RX_ACCEL_ENABLE_IP_CHECK

If an IPv4 frame is received with a mismatching header checksum, the frame is discarded. IPv6 has no header checksum and is not affected by this setting. Discarding is only available when the RX FIFO operates in store and forward mode (RSFL cleared).

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.83 Parameter ENET_RX_ACCEL_ENABLE_PROTO_CHECK

If a TCP/IP, UDP/IP, or ICMP/IP frame is received that has a wrong TCP, UDP, or ICMP checksum, the frame is discarded. Discarding is only available when the RX FIFO operates in store and forward mode (RSFL cleared).

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
defaultValue	false

4.1.84 Parameter ENET_RX_ACCEL_ENABLE_MAC_CHECK

Any frame received with a CRC, length, or PHY error is automatically discarded and not forwarded to the user application interface.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.85 Parameter ENET_RX_ACCEL_ENABLE_SHIFT16

When this field is set, the actual frame data starts at bit 16 of the first word read from the RX FIFO aligning the Ethernet payload on a 32-bit boundary. **NOTE:** This function only affects the FIFO storage and has no influence on the statistics, which use the actual length of the frame received.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.86 Container EthCtrlConfigTxAcceleratorOptions

Enable/Disable available transmit acceleration options.

Included subcontainers:

Configuration Parameters

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.87 Parameter ENET_TX_ACCEL_ENABLE_SHIFT16

Indicates to the transmit data FIFO that the written frames contain two additional octets before the frame data. This means the actual frame begins at bit 16 of the first word written into the FIFO. This function allows putting the frame payload on a 32-bit boundary in memory, as the 14-byte Ethernet header is extended to a 16-byte header.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.88 Parameter ENET_TX_ACCEL_INSERT_IP_CHECKSUM

If an IP frame is transmitted, the checksum is inserted automatically. The IP header checksum field must be cleared. If a non-IP frame is transmitted the frame is not modified.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
defaultValue	false

4.1.89 Parameter ENET_TX_ACCEL_INSERT_PROTO_CHECKSUM

If an IP frame with a known protocol is transmitted, the checksum is inserted automatically into the frame. The checksum field must be cleared. The other frames are not modified.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.1.90 Container EthCtrlConfigEgress

Controller Configuration Egress: Configuration of one Ethernet controller egress behavior.

Included subcontainers:

- [EthCtrlConfigEgressFifo](#)
- [EthCtrlConfigScheduler](#)
- [EthCtrlConfigShaper](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.91 Reference EthCtrlConfigEgressLastSchedulerRef

Reference to the scheduler which is the last in the Egress structure.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC

Configuration Parameters

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔Egress/EthCtrlConfigScheduler

4.1.92 Container EthCtrlConfigEgressFifo

Controller Configuration Egress: Represents a FIFO at the egress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.93 Parameter EthCtrlConfigEgressFifoBufLenByte

Length of FIFO elements in bytes.

Note that this represents the length of a single packet buffer (including the 14-bytes Ethernet frame header and the 4-bytes FCS).

Therefore, the payload length will be equal to EthCtrlConfigEgressFifoBufLenByte - 18.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	64
max	1536
min	64

4.1.94 Parameter EthCtrlConfigEgressFifoBufTotal

FIFO Buffer Count (defines the number of packet buffers of length EthCtrlConfigEgressFifoBufLenByte).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	16
max	32
min	1

4.1.95 Parameter EthCtrlConfigEgressFifoIdx

Egress FIFO index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	0
min	0

4.1.96 Parameter EthCtrlConfigEgressFifoCallback

Callback function invoked when an Egress (Tx) FIFO event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	Eth_TxIrqCallback

4.1.97 Parameter EthCtrlConfigEgressFifoPriorityAssignment

Message egress priority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	8
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	7
min	0

4.1.98 Container EthCtrlConfigScheduler

Controller Configuration Scheduler: Represents a Scheduler on the Egress side.

Included subcontainers:

- [EthCtrlConfigSchedulerPredecessor](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.99 Container EthCtrlConfigSchedulerPredecessor

Controller Configuration Scheduler Predecessors: Defines an ordered list of predecessors for this scheduler.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.100 Parameter EthCtrlConfigSchedulerPredecessorOrder

Defines the order of the scheduler predecessors.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.1.101 Reference EthCtrlConfigSchedulerPredecessorRef

Choice reference to the scheduler predecessor.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔Egress/EthCtrlConfigEgressFifo', '/AUTOSAR/EcucDefs/Eth/EthConfigSet/↔EthCtrlConfig/EthCtrlConfigEgress/EthCtrlConfigScheduler', '/AUTOSA↔R/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfigEgress/EthCtrl↔ConfigShaper']

4.1.102 Container EthCtrlConfigShaper

Controller Configuration Shaper : Represents a Shaper at the Egress side.

Included subcontainers:

- [EthCtrlConfigShaperVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.103 Parameter EthCtrlConfigShaperIdleSlope

Defines the increase of credit in bits per second for the AVB shaper.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	4294967295
min	0

4.1.104 Parameter EthCtrlConfigShaperMaxCredit

Maximum amount of credits that can be accumulated for a queue.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	49
min	0

4.1.105 Parameter EthCtrlConfigShaperMinCredit

Minimum amount of credits in bytes that can be accumulated for a queue.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Configuration Parameters

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	9
min	0

4.1.106 Reference EthCtrlConfigShaperPredecessorFifoRef

Reference to the FIFO which is the predecessor for this Shaper.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔Egress/EthCtrlConfigEgressFifo

4.1.107 Container EthCtrlConfigShaperVendorSpecific

Shaper vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.108 Parameter EthCtrlConfigHiCredit

Defines the maximum value (in bits) that can be accumulated in the credit parameter.

This value should be equal to the maximum size (in bits) of an interfering (best-effort) frame.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	12000
max	131072
min	0

4.1.109 Parameter EthCtrlConfigLoCredit

Defines the minimum value (in bits) that can be accumulated in the credit parameter.

This value should be equal to twice the maximum size (in bits) of a frame transmitted from this queue.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	4096
max	131072
min	0

4.1.110 Container EthCtrlConfigIngress

Controller Configuration Ingress: Configuration of one Ethernet controller ingress behavior.

Included subcontainers:

- [EthCtrlConfigIngressFifo](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Configuration Parameters

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.111 Container EthCtrlConfigIngressFifo

Controller Configuration Ingress : Represents a FIFO at the ingress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.112 Parameter EthCtrlConfigIngressFifoBufLenByte

Length of FIFO elements in bytes.

Note that this represents the length of a single packet buffer (including the 14-bytes Ethernet frame header and the 4-bytes FCS).

Therefore, the payload length will be equal to EthCtrlConfigIngressFifoBufLenByte - 18.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	1536
min	64

4.1.113 Parameter EthCtrlConfigIngressFifoBufTotal

FIFO Buffer Count (defines the number of packet buffers of length EthCtrlConfigIngressFifoBufLenByte).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16
max	32
min	1

4.1.114 Parameter EthCtrlConfigIngressFifoIdx

Ingress FIFO index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	0
min	0

4.1.115 Parameter EthCtrlConfigIngressFifoCallback

Callback function invoked when an Ingress (Rx) FIFO event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	Eth_RxIrqCallback

4.1.116 Parameter EthCtrlConfigIngressFifoPriorityAssignment

Message ingress priority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	8
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	7
min	0

4.1.117 Container EthDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs.

The EventId is taken from the referenced DemEventParameter's DemEventId value.

The standardized errors are provided in the container and can be extended by vendor specific error references.

Container is editable only when EthGeneral/EthGeneralVendorSpecific/EthDisableDemEventDetect = FALSE.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.118 Reference ETH_E_ACCESS

Reference to the DemEventParameter which shall be issued when the error 'Controller access failed' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.119 Reference ETH_E_RX_FRAMES_LOST

Reference to the DemEventParameter which shall be issued when the error 'Rx Frame Lost' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

Configuration Parameters

Property	Value
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.120 Reference ETH_E_CRC

Reference to the DemEventParameter which shall be issued when the error 'CRC Failure' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.121 Reference ETH_E_UNDERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error 'Undersized Frame' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.122 Reference ETH_E_OVERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error 'Oversized Frame' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.123 Reference ETH_E_ALIGNMENT

Reference to the DemEventParameter which shall be issued when the error 'Alignment Error' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.1.124 Reference ETH_E_SINGLECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Single Collision' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC

Configuration Parameters

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.1.125 Reference ETH_E_MULTIPLECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Multiple Collisions' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.1.126 Reference ETH_E_LATECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Late Collisions' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.1.127 Container EthCtrlConfigSpiConfiguration

SPI Interface configuration of one Ethernet controller (MACPHY use).

Included subcontainers:

- [EthCtrlConfigSpiSequence](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.128 Parameter EthCtrlConfigSpiCommTimeout

Indicates the maximum time allowed to the Ethernet controller for replying (either positively or negatively) to a SPI command. Timeout is configured in seconds. Timeout value of 0 means that no specific timeout is to be used by Ethernet controller and the communication is executed at the best of the SPI HW capacity

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
defaultValue	0.0
max	0.1
min	0.0

4.1.129 Parameter EthCtrlConfigSpiChunkPayloadSize

Configures the size of the payload chunks which will be transferred over the SPI interface. Note: The chunk is the basic element for data transaction over the SPI which can be a section of an Ethernet frame or management command. The configured value has to be a multiple of 8.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	64
min	8

4.1.130 Parameter EthCtrlConfigSpiCommRetries

Indicates the maximum number of communication retries in case of a failed SPI communication (applies both to timed out communication and to errors/NACK in the response data). If configured value is 0, no retry is allowed (communication is expected to succeed at first try).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	254
min	0

4.1.131 Parameter EthCtrlConfigSpiEnableControlDataProtection

Enables the control data protection. When set, all control data written to and read from the MACPHY will be transferred with its complement for detection of bit errors as defined in OA TC6 [26]. FALSE: Control data read/write protection is disabled (unprotected). TRUE: Control data read/write protection is enabled (protected).

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.132 Parameter EthCtrlConfigSpiEnableRxCSAlign

Configures the CSn Align Receive frame. TRUE: all received Ethernet frames data shall start at the beginning of the first receive data chunk payload following CSn assertion FALSE: received frames may begin within any receive data chunk of the transaction when this bit is clear.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.133 Parameter EthCtrlConfigSpiEnableRxCutThrough

When supported by the HW, enables the cut through mode of frame from the network to the SPI host.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.134 Parameter EthCtrlConfigSpiEnableRxZeroAlign

Configures the zero-align receive frame. TRUE: all received Ethernet frames data shall be aligned to start at the beginning of any receive data chunk payload. FALSE: Received frames may begin anywhere within the receive data chunk payload

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.135 Parameter EthCtrlConfigSpiEnableTransmitDataHdrSequence

When supported by the HW, enables the transmit data sequence monitoring. FALSE: transmit data header sequence bit monitoring disabled. TRUE: transmit data header sequence bit monitoring enabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.136 Parameter EthCtrlConfigSpiEnableTxChecksum

Configures the CSn Align Receive frame. TRUE: all received Ethernet frames data shall start at the beginning of the first receive data chunk payload following CSn assertion FALSE: received frames may begin within any receive data chunk of the transaction when this bit is clear

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.137 Parameter EthCtrlConfigSpiEnableTxCutThrough

Configures the CSn Align Receive frame. *TRUE*: all received Ethernet frames data shall start at the beginning of the first receive data chunk payload following CSn assertion *FALSE*: received frames may begin within any receive data chunk of the transaction when this bit is clear

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.138 Parameter EthCtrlConfigSpiSelectTimeStamp

When timestamp supported by the HW, selects size and format of the timestamps. *FALSE*: 32-bits timestamps *TRUE*: 64-bit timestamps

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.139 Parameter EthCtrlConfigSpiTransmitCreditThreshold

Configures the minimum of available transmit credit before the writing

IRQn is asserted. As per OA TC6, this information is notified by the

TXC field. 0 = 1 credit 1 = 4 credits 2 = 8 credits 3 = 16 credits

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	3
min	0

4.1.140 Container EthCtrlConfigSpiSequence

Container gives Ethernet controller driver information about one

SPI sequence. One SPI sequence used by Ethernet controller driver

is in exclusive use for it. No other driver is allowed to access

this sequence. Ethernet controller driver may use one sequence to

access n Ethernet controller hardware chips of the same type or n

sequences are used to access one single Ethernet controller hardware

chip. If a Ethernet controller hardware has no SPI interface, there

is no instance of this container.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.1.141 Parameter EthCtrlConfigSpiAccessSynchronous

This parameter is used to define whether the access to the Spi sequence is synchronous or asynchronous. true: SPI access is synchronous. false: SPI access is asynchronous

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.1.142 Reference EthCtrlConfigSpiSequenceName

Reference to a Spi sequence configuration container.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Spi/SpiDriver/SpiSequence

4.1.143 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.1.144 Parameter ModuleId

Module ID of this module from the BSW Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	88
max	88
min	88

4.1.145 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

4.1.146 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU,

BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name.

In total, the implementation specific name is generated as follows:

[ModuleName]_[VendorId]_[VendorApiInfix][API name from SWS].

E.g. Assuming that the VendorId of the implementor is 123 and the implementer chose a

VendorApiInfix of 'v11r456' an API named Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1.

It shall not be used for modules with upper multiplicity = 1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	ENET

4.1.147 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.1.148 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	7
max	7
min	7

4.1.149 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP

Configuration Parameters

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.150 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	3
max	3
min	3

4.1.151 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.1.152 Parameter SwPatchVersion

Patch version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

ETH Driver	102
ENET Driver	103
Ethernet Driver	145



Chapter 6

Data Structure Index

6.1 Data Structures

Here are the data structures with brief descriptions:

Enet_Ip_ManagementInfo	
Management Frame information	151



Chapter 7

Module Documentation

7.1 ETH Driver

7.1.0.1 Functions Expectation called for transmission in polling mode:

- Eth_ProvideTxBuffer
- Eth_Transmit
- Eth_TxConfirmation

7.2 ENET Driver

7.2.1 Detailed Description

7.2.1.1 Platform-specific prerequisites

None

7.2.1.2 How to use the ENET driver in your application

The ENET driver does not handle clock setup or any kind of pin configuration. This is handled by the **Clock Driver** and **Pin Driver** modules, respectively. The driver assumes that the correct clock and pins configurations have been made, so it is the user's responsibility to set up clocking and pin configurations correctly.

In order to use the ENET driver in your application, the [Enet_Ip_Init\(\)](#) function should be called prior to using the rest of the API. The parameters of this function specify:

- The ENET instance to be initialized
- A structure which will hold the internal state of the driver
- A structure specifying the configuration of the ENET module
- Two arrays of structures specifying the configuration of the buffers (one structure for each rings - an array for Rx and one for Tx)
- The MAC address to be configured for the module

The configuration of the module is specified through the [Enet_Ip_ConfigType](#) structure and contains:

- MII-related configurations (mode, speed, duplex)
- Configuration options for the receive and transmit path
- The maximum frame length
- The MAC interrupt sources which should be enabled
- A callback function to be invoked on events
- The number of Rx and Tx rings used by the driver
- MAC address filtering options

The buffers configuration is specified through the [Enet_Ip_BufferConfigType](#) structure and contains:

- The size of the ring
- The size of each buffer in ring
- Pointers to the beginning of the buffer descriptor rings
- Pointers to the beginning of the memory area where the buffer data resides
- Channel interrupt sources
- A callback function to be invoked on per channel events

In order to de-initialize the driver, the [Enet_Ip_Deinit\(\)](#) function shall be used. This function will disable the ENET interrupts and the module, so calling other ENET driver functions after de-initializing the driver will have undefined behavior. In order to use the driver again, [Enet_Ip_Init\(\)](#) should be called.

7.2.1.3 How to enable zero-copy optimization

The buffers allocation is controlled by the following configuration parameters:

- EthCtrlAllocateTxDataBuffers
- EthCtrlAllocateRxDataBuffers

Setting them both to FALSE will prevent the driver from using internal buffers.

However, it is important to keep in mind that the programming sequences will also be slightly affected:

- EthAllocateTxDataBuffers = TRUE
 - Call sequence for each frame: Enet_Ip_GetTxBuff -> Enet_Ip_SendFrame
 - Enet_Ip_SendFrame must be called with the same buffer returned by Enet_Ip_GetTxBuff
- EthAllocateTxDataBuffers = FALSE
 - Call sequence for each frame: Enet_Ip_SendFrame
 - Enet_Ip_SendFrame can be called with any buffer provided by the application
- EthCtrlAllocateRxDataBuffers = TRUE
 - Call sequence for each frame: Enet_Ip_ReadFrame -> Enet_Ip_ProvideRxBuff
 - Enet_Ip_ProvideRxBuff must be called with the same buffer returned by Enet_Ip_ReadFrame
- EthCtrlAllocateRxDataBuffers = FALSE
 1. Application has an RX buffer pool
 - After controller initialization: Call Enet_Ip_ProvideRxBuff for each buffer
 - Call sequence for each frame: Enet_Ip_ReadFrame -> Enet_Ip_ProvideRxBuff
 - Enet_Ip_ProvideRxBuff can be called with either:
 - * The same buffer returned by Enet_Ip_ReadFrame if the application wants to reuse the buffer
 - * A buffer different than the one returned by Enet_Ip_ReadFrame if the application doesn't want to reuse the buffer
 2. Application doesn't have an RX buffer pool and wants to provide RX buffers on demand
 - After controller initialization: Nothing to be done
 - Call sequence for each frame: Enet_Ip_ProvideRxBuff -> Enet_Ip_ReadFrame

7.2.1.4 How to calculate size of ethernet buffer areas

Ethernet buffer areas is placed in noncache memory for each queue in each controller with order:

- Rx descriptor buffer areas: number of buffers configured(by EthCtrlConfigIngressFifoBufTotal) * size of each buffer ([Enet_Ip_BufferDescriptorType](#)).
- Rx data buffer areas: number of buffers configured(by EthCtrlConfigIngressFifoBufTotal) * size of each buffer (EthCtrlConfigIngressFifoBufLenByte).
- Tx descriptor buffer areas: number of buffers configured(by EthCtrlConfigEgressFifoBufTotal) * size of each buffer ([Enet_Ip_BufferDescriptorType](#)).
- Tx data buffer areas: number of buffers configured(by EthCtrlConfigEgressFifoBufTotal) * size of each buffer (EthCtrlConfigEgressFifoBufLenByte).

All of above memory areas need to align with start address by `FEATURE_ENET_BUFFDESCR_ALIGNMENT_BYTES(16U)`.

This configuration will be placed in loop with multiple queues and multiple controllers.

In code:

- For Reception buffer descriptors for Rx Ring `["num:i($RingIdx)"]` :

```
VAR_ALIGN(Enet_Ip_BufferDescriptorType ENET_["num:i($CtrlIdx)"]_RxRing_["num:i($RingIdx)"]_DescBuffer[ENET_["num:i($CtrlIdx)"]_
FEATURE_ENET_BUFFDESCR_ALIGNMENT_BYTES])
```

- For Reception data buffers for Rx Ring `["num:i($RingIdx)"]`

```
VAR_ALIGN(uint8 ENET_["num:i($CtrlIdx)"]_RxRing_["num:i($RingIdx)"]_DataBuffer[ENET_["num:i($CtrlIdx)"]_MAX_RXBUFF_SUPPORTED]
* ENET_["num:i($CtrlIdx)"]_MAX_RXBUFFLEN_SUPPORTED], FEATURE_ENET_BUFFDATA_ALIGNMENT_BYTES)
```

- For Transmission buffer descriptors for Tx Ring `["num:i($RingIdx)"]`

```
VAR_ALIGN(Enet_Ip_BufferDescriptorType ENET_["num:i($CtrlIdx)"]_TxRing_["num:i($RingIdx)"]_DescBuffer[ENET_["num:i($CtrlIdx)"]_
FEATURE_ENET_BUFFDESCR_ALIGNMENT_BYTES])
```

- For Transmission data buffers for Tx Ring `["num:i($RingIdx)"]`

```
VAR_ALIGN(uint8 ENET_["num:i($CtrlIdx)"]_TxRing_["num:i($RingIdx)"]_DataBuffer[ENET_["num:i($CtrlIdx)"]_MAX_TXBUFF_SUPPORTED]
* ENET_["num:i($CtrlIdx)"]_MAX_TXBUFFLEN_SUPPORTED], FEATURE_ENET_BUFFDATA_ALIGNMENT_BYTES)
```

7.2.1.5 How to set up MAC address filters

The Address Filtering Module of the MAC checks the source address and destination address fields of each incoming packet. By default, the module only receives the frames having the destination address equal to the MAC address with which the ENET module has been initialized or the broadcast address.

The module has 1 additional 48-bit perfect address filter with masks for each byte, which can be configured for either source address or destination address filtering. This filter can be set-up by calling the `ENET_Ip_SetAddrPerfectFilter()` function.

Multicast and unicast destination addresses can be added to the hash filter by calling the `Enet_Ip_AddDstAddrToHashFilter()` function and removed by calling the `Enet_Ip_RemoveDstAddrFromHashFilter()` function.

For multicast and broadcast frames, the filters can be bypassed by calling the `Enet_Ip_SetBroadcastForwardAll()` and `Enet_Ip_SetMulticastForwardAll()` functions.

7.2.1.6 Examples:

Initializing the module:

```
Enet_Ip_Init(INST_ENET_0, &ENET_0_ConfigPB_<functionalGroupName>);
```

```
/* ... */
```

```
Enet_Ip_Deinit(INST_ENET_0);
```

Sending a frame:

```

Enet_lp_BufferType buffer;
Enet_lp_TxOptionsType options = {FALSE, FALSE, ENET_CHECKSUM_INSERTION_DISABLE};

buffer.length = 8;
if (ENET_STATUS_SUCCESS == Enet_lp_GetTxBuff(INST_ENET_0, 0, &buffer, NULL_PTR))
{
    for (uint32 i = 0; i < 8; ++i)
    {
        buffer.data[i] = i;
    }

    Enet_lp_SendFrame(INST_ENET_0, 0, &buffer, &options);
}

```

Receiving a frame - polling method (EthCtrlEnableRxInterrupt = FALSE):

```

Enet_lp_BufferType buffer;
Enet_lp_RxInfoType info;

for (;;)
{
    if (status == Enet_lp_ReadFrame(INST_ENET_0, 0, &buffer, &info))
    {
        /* Process buffer */

        /* buffer is no longer needed, provide it to the driver in order to be used by the reception mechanism */
        Enet_lp_ProvideRxBuff(INST_ENET_0, 0, &buffer);
    }
}

```

Receiving a frame - interrupt method (EthCtrlEnableRxInterrupt = TRUE):

```

void rx_callback(uint8 instance, uint8 channel)
{
    Enet_lp_BufferType buffer;

    if (status == Enet_lp_ReadFrame(INST_ENET_0, 0, &buffer, NULL_PTR))
    {
        /* Process buffer */

        /* buffer is no longer needed, provide it to the driver in order to be used by the reception mechanism */
        Enet_lp_ProvideRxBuff(INST_ENET_0, 0, &buffer);
    }
}

```



```

}
}

int main(void)
{
    /* ... */

    Enet_Ip_Init(INST_ENET_0, &ENET_0_ConfigPB_<functionalGroupName>);

    /* ... */
}

```

Data Structures

- struct [Enet_Ip_BufferType](#)
Send/Receive buffer information for the user. [More...](#)
- struct [Enet_Ip_BufferDescriptorType](#)
Send/Receive internal buffer descriptor. [More...](#)
- struct [Enet_Ip_BufferConfigType](#)
Specialization of ring configuration for Tx Rings. [More...](#)
- struct [Enet_Ip_ConfigType](#)
Defines the ENET module configuration structure. [More...](#)
- struct [Enet_Ip_StateType](#)
Internal driver state structure. [More...](#)
- struct [Enet_Ip_TimeStampType](#)
Defines the TimeStamp. [More...](#)
- struct [Enet_Ip_RxInfoType](#)
Enhanced information related to a receive buffer descriptor. [More...](#)
- struct [Enet_Ip_TxInfoType](#)
Enhanced information related to a transmit buffer descriptor. [More...](#)
- struct [Enet_Ip_TimerConfigType](#)
Configuration of the adjustable timer. [More...](#)
- struct [Enet_Ip_TimerChannelConfigType](#)
Configuration structure of the timer channels. [More...](#)
- struct [Enet_Ip_TxOptionsType](#)
Transmit options for a particular frame. [More...](#)

Types Reference

- typedef void(* [Enet_Ip_CallbackType](#)) (uint8 Instance, [Enet_Ip_EventType](#) Event)
Callback function invoked when a general event is encountered.
- typedef void(* [Enet_Ip_ChCallbackType](#)) (uint8 Instance, uint8 Ring)
Callback function invoked when a channel event is encountered.

- typedef void(* [Enet_Ip_TimerCallbackType](#)) (uint8 Instance)

Callback function invoked when a timer event is encountered. The channels parameter represents a mask of the channels on which an event occurred.

Enum Reference

- enum [Enet_Ip_StatusType](#)
Return values.
- enum [Enet_Ip_EventType](#)
Event specifier for the callback function.
- enum [Enet_Ip_TxSpecialConfigType](#)
Special transmit control configurations.
- enum [Enet_Ip_TxAcceleratorType](#)
Transmit accelerator configurations.
- enum [Enet_Ip_RxAcceleratorType](#)
Receive accelerator configurations.
- enum [Enet_Ip_CounterType](#)
Statistics counters enumeration.
- enum [Enet_Ip_MiiSpeedType](#)
Media Independent Interface speed selection.
- enum [Enet_Ip_PowerStateType](#)
Controller power state indication.
- enum [Enet_Ip_SysTimeCorrOffsetType](#)
System time correction offset.
- enum [Enet_Ip_TimerChannelModeType](#)
Timer channels configuration modes.
- enum [Enet_Ip_ChecksumInsControlType](#)
Checksum insertion control.
- enum [Enet_Ip_MmfrOpType](#)
Management Frame operation type.
- enum [Enet_Ip_FifoThresholdType](#)
Available FIFO threshold levels.

Initialization and De-initialization

- void [Enet_Ip_Init](#) (uint8 Instance, const Enet_CtrlConfigType *Config)
Initializes the ENET module.
- [Enet_Ip_PowerStateType](#) [Enet_Ip_GetPowerState](#) (uint8 Instance)
Gets the current power state of the ENET module.
- void [Enet_Ip_Deinit](#) (uint8 Instance)
Deinitializes the ENET module.
- void [Enet_Ip_EnableController](#) (uint8 Instance)
Enables all configured transmit and receive buffers and then enables the controller.

- void [Enet_Ip_RecoverFilter](#) (uint8 Instance)
Recover the hash table filtering caused by add add/remove open/close to hash table.
- void [Enet_Ip_DisableController](#) (uint8 Instance)
Disables the controller and resets all the configured transmit and receive buffers.
- void [Enet_Ip_SetSpeed](#) (uint8 Instance, [Enet_Ip_MiiSpeedType](#) Speed)
Sets the speed of the MII interface.
- [Enet_Ip_StatusType](#) [Enet_Ip_GetTxBuff](#) (uint8 Instance, uint8 Ring, [Enet_Ip_BufferType](#) *Buff, uint16 *BuffId)
Provides a transmit buffer to be used by the application for transmission.
- [Enet_Ip_StatusType](#) [Enet_Ip_SendFrame](#) (uint8 Instance, uint8 Ring, const [Enet_Ip_BufferType](#) *Buff, const [Enet_Ip_TxOptionsType](#) *Options)
Sends an Ethernet frame.
- [Enet_Ip_StatusType](#) [Enet_Ip_ReadFrame](#) (uint8 Instance, uint8 Ring, [Enet_Ip_BufferType](#) *Buff, [Enet_Ip_RxInfoType](#) *Info)
Reads a received Ethernet frame.
- void [Enet_Ip_ProvideRxBuff](#) (uint8 Instance, uint8 Ring, const [Enet_Ip_BufferType](#) *Buff)
Provides a receive buffer to be used by the driver for reception.
- boolean [Enet_Ip_IsFrameAvailable](#) (uint8 Instance, uint8 Ring)
Checks if there are more frames available in the given queue.
- [Enet_Ip_StatusType](#) [Enet_Ip_GetTransmitStatus](#) (uint8 Instance, uint8 Ring, const [Enet_Ip_BufferType](#) *Buff, [Enet_Ip_TxInfoType](#) *Info)
Checks if the transmission of a buffer is complete.
- uint32 [Enet_Ip_GetCounter](#) (uint8 Instance, [Enet_Ip_CounterType](#) Counter)
Gets statistics from the specified counter.
- void [Enet_Ip_ConfigCounters](#) (uint8 Instance, boolean Enable)
Enables/Disables the MIB counters.
- void [Enet_Ip_EnableMDIO](#) (uint8 Instance, boolean MiiPreambleDisabled, uint32 ModuleClk)
Enables the MDIO interface.
- [Enet_Ip_StatusType](#) [Enet_Ip_MDIORead](#) (uint8 Instance, uint8 PhyAddr, uint8 PhyReg, uint16 *Data, uint32 TimeoutMs)
Reads the selected register of the PHY.
- [Enet_Ip_StatusType](#) [Enet_Ip_MDIOWrite](#) (uint8 Instance, uint8 PhyAddr, uint8 PhyReg, uint16 Data, uint32 TimeoutMs)
Writes the selected register of the PHY.
- [Enet_Ip_StatusType](#) [Enet_Ip_MDIOReadMMD](#) (uint8 Instance, uint8 PhyAddr, uint8 Mmd, uint16 Phy←Reg, uint16 *Data, uint32 TimeoutMs)
Reads a register of the specified MMD in a PHY device.
- [Enet_Ip_StatusType](#) [Enet_Ip_MDIOWriteMMD](#) (uint8 Instance, uint8 PhyAddr, uint8 Mmd, uint16 Phy←Reg, uint16 Data, uint32 TimeoutMs)
Writes a register of the specified MMD in a PHY device.
- void [Enet_Ip_SetMacAddr](#) (uint8 Instance, const uint8 *MacAddr)
Configures the physical address of the MAC.
- void [Enet_Ip_GetMacAddr](#) (uint8 Instance, uint8 *MacAddr)

Gets the physical address of the MAC.

- void [Enet_Ip_SetBroadcastForwardAll](#) (uint8 Instance, boolean Enable)

Enables/Disables forwarding of the broadcast traffic.

- void [Enet_Ip_SetMulticastForwardAll](#) (uint8 Instance, boolean Enable)

Enables/Disables forwarding of the multicast traffic, irrespective of the destination MAC address.

- void [Enet_Ip_AddDstAddrToHashFilter](#) (uint8 Instance, const uint8 *MacAddr)

Adds a hardware address to the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

- void [Enet_Ip_RemoveDstAddrFromHashFilter](#) (uint8 Instance, const uint8 *MacAddr)

Removes a hardware address from the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

- void [Enet_Ip_SetSysTimeCorr](#) (uint8 Instance, [Enet_Ip_SysTimeCorrOffsetType](#) Offset, [Enet_Ip_TimeStampType](#) DiffTimeStamp)

Set system time correction.

- void [Enet_Ip_TimerSetCorrection](#) (uint8 Instance, uint32 CorrectValue, uint32 CorrectPeriod)

Sets the timer correction period and correction increment value.

- void [Enet_Ip_TimerStop](#) (uint8 Instance)

The timer stop at the current value.

- [Enet_Ip_StatusType](#) [Enet_Ip_TimerEnableChannel](#) (uint8 Instance, uint8 Channel, const [Enet_Ip_TimerChannelConfig](#) *Config)

Configures and enables a specific timer channel.

- void [Enet_Ip_TimerSetCompare](#) (uint8 Instance, uint8 Channel, uint32 Value)

Sets the compare value for a channel configured as output compare.

- void [Enet_Ip_TimerStart](#) (uint8 Instance)

The timer starts incrementing.

- void [Enet_Ip_TimerInit](#) (uint8 Instance, const [Enet_Ip_TimerConfigType](#) *TimerConfig)

Initializes the adjustable timer using the given configuration and enables it.

- void [Enet_Ip_TimerSet](#) (uint8 Instance, [Enet_Ip_TimeStampType](#) TimerValue)

Sets the value of the timer.

- void [Enet_Ip_TimerGet](#) (uint8 Instance, [Enet_Ip_TimeStampType](#) *TimeStamp)

Gets the value of the timer.

- void [Enet_Ip_EnableTxStoreAndForward](#) (uint8 Instance)

Enable store and forward .

- void [Enet_Ip_SetFIFOThreshold](#) (uint8 Instance, [Enet_Ip_FifoThresholdType](#) ThresholdType, uint8 ThresholdValue)

Sets FIFO threshold levels.

- void [Enet_Ip_SetTxFIFOWatermark](#) (uint8 Instance, uint16 Watermark)

Sets the transmit FIFO watermark.

- #define **ETH_43_ENET_STOP_SEC_CODE**

7.2.2 Data Structure Documentation

7.2.2.1 struct Enet_Ip_BufferType

Send/Receive buffer information for the user.

Definition at line 345 of file Enet_Ip_Types.h.

7.2.2.2 struct Enet_Ip_BufferDescriptorType

Send/Receive internal buffer descriptor.

Definition at line 355 of file Enet_Ip_Types.h.

Data Fields

- uint16 [Length](#)
- uint16 [Control](#)
- uint8 * [Buffer](#)
- uint32 [Enh1](#)
- uint32 [Enh2](#)
- uint32 [Enh3](#)
- uint32 [TimeStamp](#)
- uint32 [Reserved1](#)
- uint32 [Reserved2](#)

Field Documentation

Length

uint16 Length

Buffer descriptor data length.

Definition at line 357 of file Enet_Ip_Types.h.

Control

uint16 Control

Buffer descriptor control and status.

Definition at line 358 of file Enet_Ip_Types.h.

Buffer

uint8* Buffer

Data buffer pointer.

Definition at line 359 of file Enet_Ip_Types.h.

Enh1

uint32 Enh1

First word of the extended part of the buffer descriptor.

Definition at line 360 of file Enet_Ip_Types.h.

Enh2

uint32 Enh2

Second word of the extended part of the buffer descriptor.

Definition at line 361 of file Enet_Ip_Types.h.

Enh3

uint32 Enh3

Third word of the extended part of the buffer descriptor.

Definition at line 362 of file Enet_Ip_Types.h.

TimeStamp

uint32 TimeStamp

TimeStamp of the frame.

Definition at line 363 of file Enet_Ip_Types.h.

Reserved1

uint32 Reserved1

Reserved.

Definition at line 364 of file Enet_Ip_Types.h.

Reserved2

uint32 Reserved2

Reserved.

Definition at line 365 of file Enet_Ip_Types.h.

7.2.2.3 struct Enet_Ip_BufferConfigType

Specialization of ring configuration for Tx Rings.

Definition at line 372 of file Enet_Ip_Types.h.

Data Fields

- [Enet_Ip_BufferDescriptorType](#) * [RingDesc](#)
- [Enet_Ip_ChCallbackType](#) [Callback](#)
- uint8 * [Buffer](#)
- uint32 [Interrupts](#)
- uint16 [BufferLen](#)
- uint16 [RingSize](#)

Field Documentation**RingDesc**

[Enet_Ip_BufferDescriptorType](#)* [RingDesc](#)

Buffer descriptor ring start address.

Definition at line 374 of file [Enet_Ip_Types.h](#).

Callback

[Enet_Ip_ChCallbackType](#) [Callback](#)

Callback function for current channel.

Definition at line 375 of file [Enet_Ip_Types.h](#).

Buffer

uint8* [Buffer](#)

Buffer data pool start address

Definition at line 376 of file [Enet_Ip_Types.h](#).

Interrupts

uint32 [Interrupts](#)

Channel interrupt sources. A logical OR of "Enet_Ip_InterruptType".

Definition at line 377 of file [Enet_Ip_Types.h](#).

BufferLen

uint16 [BufferLen](#)

Length of each individual buffer in a pool

Definition at line 378 of file [Enet_Ip_Types.h](#).

RingSize

uint16 RingSize

Buffer descriptors number.

Definition at line 379 of file Enet_Ip_Types.h.

7.2.2.4 struct Enet_Ip_ConfigType

Defines the ENET module configuration structure.

Definition at line 386 of file Enet_Ip_Types.h.

Data Fields

- uint8 [RxRingCount](#)
- uint8 [TxRingCount](#)
- [Enet_Ip_TimerCallbackType](#) TimerCallback
- uint32 [Interrupts](#)
- [Enet_Ip_CallbackType](#) Callback
- uint16 [MaxFameLen](#)
- [Enet_Ip_MiiModeType](#) MiiMode
- [Enet_Ip_MiiSpeedType](#) MiiSpeed
- [Enet_Ip_DuplexType](#) MiiDuplex
- uint32 [RxConfig](#)
- uint32 [TxConfig](#)
- uint8 [RxAccelerConfig](#)
- uint8 [TxAccelerConfig](#)
- boolean [EnableCtrl](#)

Field Documentation

RxRingCount

uint8 RxRingCount

The number of Receive rings to be used by the driver.

Definition at line 388 of file Enet_Ip_Types.h.

TxRingCount

uint8 TxRingCount

The number of Transmit rings to be used by the driver.

Definition at line 389 of file Enet_Ip_Types.h.

TimerCallback

`Enet_Ip_TimerCallbackType` TimerCallback

Callback to be invoked when a timer event occurs.

Definition at line 390 of file `Enet_Ip_Types.h`.

Interrupts

`uint32` Interrupts

MAC interrupt source. A logical OR of "Enet_Ip_InterruptType".

Definition at line 391 of file `Enet_Ip_Types.h`.

Callback

`Enet_Ip_CallbackType` Callback

Callback function

Definition at line 392 of file `Enet_Ip_Types.h`.

MaxFameLen

`uint16` MaxFameLen

Maximum frame length.

Definition at line 393 of file `Enet_Ip_Types.h`.

MiiMode

`Enet_Ip_MiiModeType` MiiMode

MII mode.

Definition at line 394 of file `Enet_Ip_Types.h`.

MiiSpeed

`Enet_Ip_MiiSpeedType` MiiSpeed

MII Speed.

Definition at line 395 of file `Enet_Ip_Types.h`.

MiiDuplex

Enet_Ip_DuplexType MiiDuplex

MII duplex.

Definition at line 396 of file Enet_Ip_Types.h.

RxConfig

uint32 RxConfig

MAC receive special configuration. A logical OR of "Enet_Ip_RxSpecialConfigType".

Definition at line 397 of file Enet_Ip_Types.h.

TxConfig

uint32 TxConfig

MAC transmit special configuration. A logical OR of "Enet_Ip_TxSpecialConfigType".

Definition at line 398 of file Enet_Ip_Types.h.

RxAccelerConfig

uint8 RxAccelerConfig

Receive accelerator, A logical OR of "Enet_Ip_RxAcceleratorType".

Definition at line 399 of file Enet_Ip_Types.h.

TxAccelerConfig

uint8 TxAccelerConfig

Transmit accelerator, A logical OR of "Enet_Ip_TxAcceleratorType".

Definition at line 400 of file Enet_Ip_Types.h.

EnableCtrl

boolean EnableCtrl

Configures whether the controller is enabled at initialization.

Definition at line 401 of file Enet_Ip_Types.h.

7.2.2.5 struct Enet_Ip_StateType

Internal driver state structure.

Definition at line 408 of file Enet_Ip_Types.h.

Data Fields

- [Enet_Ip_CallbackType](#) Callback
- [Enet_Ip_ChCallbackType](#) RxChCallback [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_ChCallbackType](#) TxChCallback [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_TimerCallbackType](#) TimerCallback
- [Enet_Ip_BufferDescriptorType](#) * [RxBdBase](#) [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_BufferDescriptorType](#) * [RxCurrentDesc](#) [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_BufferDescriptorType](#) * [RxBdAlloc](#) [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_BufferDescriptorType](#) * [TxBdBase](#) [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_BufferDescriptorType](#) * [TxCurrentDesc](#) [FEATURE_ENET_NUM_RINGS]
- [Enet_Ip_BufferDescriptorType](#) * [TxTransmitDesc](#) [FEATURE_ENET_NUM_RINGS]
- `uint8 * TxDataBuffer` [FEATURE_ENET_NUM_RINGS]
- `uint8 * RxDataBuffer` [FEATURE_ENET_NUM_RINGS]
- `uint16 RxBuffNums` [FEATURE_ENET_NUM_RINGS]
- `uint16 TxBuffNums` [FEATURE_ENET_NUM_RINGS]
- `uint16 RxBuffLen` [FEATURE_ENET_NUM_RINGS]
- `uint16 TxBuffLen` [FEATURE_ENET_NUM_RINGS]
- `uint8 RxRingCount`
- `uint8 TxRingCount`

Field Documentation**Callback**

[Enet_Ip_CallbackType](#) Callback

Callback function for general interrupts.

Definition at line 410 of file [Enet_Ip_Types.h](#).

RxChCallback

[Enet_Ip_ChCallbackType](#) RxChCallback [FEATURE_ENET_NUM_RINGS]

Callback function for receive channels.

Definition at line 411 of file [Enet_Ip_Types.h](#).

TxChCallback

[Enet_Ip_ChCallbackType](#) TxChCallback [FEATURE_ENET_NUM_RINGS]

Callback function for transmit channels.

Definition at line 412 of file [Enet_Ip_Types.h](#).

TimerCallback

`Enet_Ip_TimerCallbackType` TimerCallback

Timer callback function.

Definition at line 413 of file `Enet_Ip_Types.h`.

RxBdBase

`Enet_Ip_BufferDescriptorType*` RxBdBase[FEATURE_ENET_NUM_RINGS]

Receive buffer descriptor base address pointer array.

Definition at line 414 of file `Enet_Ip_Types.h`.

RxCurrentDesc

`Enet_Ip_BufferDescriptorType*` RxCurrentDesc[FEATURE_ENET_NUM_RINGS]

The current available receive buffer descriptor pointer array.

Definition at line 415 of file `Enet_Ip_Types.h`.

RxBdAlloc

`Enet_Ip_BufferDescriptorType*` RxBdAlloc[FEATURE_ENET_NUM_RINGS]

Pointer array used for enqueueing Rx buffers provided using `ENET_DRV_ProvideRxBuff`.

Definition at line 416 of file `Enet_Ip_Types.h`.

TxBdBase

`Enet_Ip_BufferDescriptorType*` TxBdBase[FEATURE_ENET_NUM_RINGS]

Transmit buffer descriptor base address pointer array.

Definition at line 417 of file `Enet_Ip_Types.h`.

TxCurrentDesc

`Enet_Ip_BufferDescriptorType*` TxCurrentDesc[FEATURE_ENET_NUM_RINGS]

The current available provided buffer descriptor pointer array.

Definition at line 418 of file `Enet_Ip_Types.h`.

TxTransmitDesc

```
Enet_Ip_BufferDescriptorType* TxTransmitDesc[FEATURE_ENET_NUM_RINGS]
```

The current available transmit buffer descriptor pointer array.

Definition at line 419 of file Enet_Ip_Types.h.

TxDataBuffer

```
uint8* TxDataBuffer[FEATURE_ENET_NUM_RINGS]
```

Transmit data buffer base address pointer array.

Definition at line 420 of file Enet_Ip_Types.h.

RxDataBuffer

```
uint8* RxDataBuffer[FEATURE_ENET_NUM_RINGS]
```

Receive data buffer base address pointer array.

Definition at line 421 of file Enet_Ip_Types.h.

RxBuffNums

```
uint16 RxBuffNums[FEATURE_ENET_NUM_RINGS]
```

Number of Rx buffers in ring

Definition at line 422 of file Enet_Ip_Types.h.

TxBuffNums

```
uint16 TxBuffNums[FEATURE_ENET_NUM_RINGS]
```

Number of Tx buffers in ring

Definition at line 423 of file Enet_Ip_Types.h.

RxBuffLen

```
uint16 RxBuffLen[FEATURE_ENET_NUM_RINGS]
```

Rx Length of buffer

Definition at line 424 of file Enet_Ip_Types.h.

TxBuffLen

uint16 TxBuffLen[FEATURE_ENET_NUM_RINGS]

Tx Length of buffer

Definition at line 425 of file Enet_Ip_Types.h.

RxRingCount

uint8 RxRingCount

The number of rings used by the driver.

Definition at line 426 of file Enet_Ip_Types.h.

TxRingCount

uint8 TxRingCount

The number of rings used by the driver.

Definition at line 427 of file Enet_Ip_Types.h.

7.2.2.6 struct Enet_Ip_TimeStampType

Defines the TimeStamp.

Definition at line 444 of file Enet_Ip_Types.h.

Data Fields

- uint32 [nanoseconds](#)
- uint32 [seconds](#)
- uint16 [secondsHi](#)

Field Documentation

nanoseconds

uint32 nanoseconds

Nanoseconds part of the time.

Definition at line 446 of file Enet_Ip_Types.h.

seconds

uint32 seconds

The 32 least significant bits of the 48 bits seconds part of the time.

Definition at line 447 of file Enet_Ip_Types.h.

secondsHi

uint16 secondsHi

The 16 most significant bits of the 48 bit seconds part of the time.

Definition at line 448 of file Enet_Ip_Types.h.

7.2.2.7 struct Enet_Ip_RxInfoType

Enhanced information related to a receive buffer descriptor.

Definition at line 455 of file Enet_Ip_Types.h.

Data Fields

- uint32 [ErrMask](#)
- uint16 [Checksum](#)
- uint8 [VlanPrio](#)
- uint8 [HeaderLen](#)
- uint16 [PktLen](#)
- uint8 [Proto](#)
- boolean [Unicast](#)
- boolean [Ipv6](#)
- boolean [Ipv4Frag](#)
- [Enet_Ip_TimeStampType](#) [TimeStamp](#)

Field Documentation**ErrMask**

uint32 ErrMask

Mask of ENET_RX_ENH_ERR_*, representing the errors that occurred.

Definition at line 456 of file Enet_Ip_Types.h.

Checksum

uint16 Checksum

One's complement sum of the payload section of the IP frame.

Definition at line 457 of file Enet_Ip_Types.h.

VlanPrio

uint8 VlanPrio

Frame priority level. Valid values are from 0 (best effort) to 7 (highest).

Definition at line 458 of file Enet_Ip_Types.h.

HeaderLen

uint8 HeaderLen

Number of 32-bit words found within the IP and its following protocol headers.

Definition at line 459 of file Enet_Ip_Types.h.

PktLen

uint16 PktLen

Byte length of the received packet that was transferred to system memory (including CRC).

Definition at line 460 of file Enet_Ip_Types.h.

Proto

uint8 Proto

The 8-bit protocol field found within the IP header of the frame.

Definition at line 461 of file Enet_Ip_Types.h.

Unicast

boolean Unicast

Indicates that the frame is an unicast frame.

Definition at line 462 of file Enet_Ip_Types.h.

Ipv6

boolean Ipv6

Indicates that the frame is an IPv6 frame.

Definition at line 463 of file Enet_Ip_Types.h.

Ipv4Frag

boolean Ipv4Frag

Indicates that the frame is an IPv4 fragment frame.

Definition at line 464 of file Enet_Ip_Types.h.

TimeStamp

[Enet_Ip_TimeStampType](#) TimeStamp

TimeStamp of the received frame.

Definition at line 465 of file Enet_Ip_Types.h.

7.2.2.8 struct Enet_Ip_TxInfoType

Enhanced information related to a transmit buffer descriptor.

Definition at line 472 of file Enet_Ip_Types.h.

Data Fields

- uint32 [ErrMask](#)
- [Enet_Ip_TimeStampType](#) TimeStamp

Field Documentation

ErrMask

uint32 ErrMask

Mask of ENET_TX_ENH_ERR_*, representing the errors that occurred.

Definition at line 473 of file Enet_Ip_Types.h.

TimeStamp

[Enet_Ip_TimeStampType](#) TimeStamp

TimeStamp of the transmitted frame.

Definition at line 474 of file Enet_Ip_Types.h.

7.2.2.9 struct Enet_Ip_TimerConfigType

Configuration of the adjustable timer.

Definition at line 481 of file Enet_Ip_Types.h.

Data Fields

- uint32 [TimerPeriod](#)
- uint8 [TimerInc](#)
- uint32 [CorrectionPeriod](#)
- uint8 [CorrectionInc](#)

Field Documentation

TimerPeriod

uint32 TimerPeriod

When the timer reaches this value, the period event occurs and the timer restarts.

Definition at line 482 of file Enet_Ip_Types.h.

TimerInc

uint8 TimerInc

The timer increments by this amount each clock cycle.

Definition at line 483 of file Enet_Ip_Types.h.

CorrectionPeriod

uint32 CorrectionPeriod

Defines after how many timer clock cycles the correction counter should be reset and triggers a correction increment on the timer.

Definition at line 484 of file Enet_Ip_Types.h.

CorrectionInc

uint8 CorrectionInc

This value is added every time the correction timer expires.

Definition at line 486 of file Enet_Ip_Types.h.

7.2.2.10 struct Enet_Ip_TimerChannelConfigType

Configuration structure of the timer channels.

Definition at line 517 of file Enet_Ip_Types.h.

Data Fields

- [Enet_Ip_TimerChannelModeType](#) Mode
- boolean [IrqEnable](#)
- boolean [DmaEnable](#)

Field Documentation

Mode

[Enet_Ip_TimerChannelModeType](#) Mode

The mode in which the timer channel shall be configured.

Definition at line 518 of file Enet_Ip_Types.h.

IrqEnable

boolean IrqEnable

Enable timer interrupt for this channel. Note: When enabling a channel interrupt, the timer interrupt must also be enabled.

Definition at line 519 of file Enet_Ip_Types.h.

DmaEnable

boolean DmaEnable

Enable DMA request for this channel.

Definition at line 521 of file Enet_Ip_Types.h.

7.2.2.11 struct Enet_Ip_TxOptionsType

Transmit options for a particular frame.

Definition at line 552 of file Enet_Ip_Types.h.

Data Fields

- boolean [NoCRC](#)
- boolean [NoInt](#)
- [Enet_Ip_ChecksumInsControlType](#) [ChecksumIns](#)

Field Documentation**NoCRC**

boolean NoCRC

Do not append CRC. It will be provided by the application.

Definition at line 553 of file Enet_Ip_Types.h.

NoInt

boolean NoInt

Do not generate a transmit interrupt.

Definition at line 554 of file Enet_Ip_Types.h.

ChecksumIns

[Enet_Ip_ChecksumInsControlType](#) [ChecksumIns](#)

Controls the IP checksum insertion for Tx packet.

Definition at line 555 of file Enet_Ip_Types.h.

7.2.3 Types Reference

7.2.3.1 Enet_Ip_CallbackType

```
typedef void(* Enet_Ip_CallbackType) (uint8 Instance, Enet_Ip_EventType Event)
```

Callback function invoked when a general event is encountered.

Definition at line 217 of file Enet_Ip_Types.h.

7.2.3.2 Enet_Ip_ChCallbackType

```
typedef void(* Enet_Ip_ChCallbackType) (uint8 Instance, uint8 Ring)
```

Callback function invoked when a channel event is encountered.

Definition at line 222 of file Enet_Ip_Types.h.

7.2.3.3 Enet_Ip_TimerCallbackType

```
typedef void(* Enet_Ip_TimerCallbackType) (uint8 Instance)
```

Callback function invoked when a timer event is encountered. The channels parameter represents a mask of the channels on which an event occurred.

Implements : Enet_Ip_TimerCallbackType_class

Definition at line 230 of file Enet_Ip_Types.h.

7.2.4 Enum Reference

7.2.4.1 Enet_Ip_StatusType

```
enum Enet_Ip_StatusType
```

Return values.

Enumerator

ENET_STATUS_SUCCESS	Generic operation success status
ENET_STATUS_ERROR	Generic operation failure status
ENET_STATUS_BUSY	Generic operation busy status
ENET_STATUS_TIMEOUT	Generic operation timeout status
ENET_STATUS_UNSUPPORTED	Generic operation unsupported status
ENET_STATUS_CACHE_ERROR	Generic operation cache operation failed
ENET_STATUS_RX_QUEUE_EMPTY	There is no available frame in the receive queue
ENET_STATUS_TX_QUEUE_FULL	There is no available space for the frame in the transmit queue
ENET_STATUS_BUFF_NOT_FOUND	The specified buffer was not found in the queue
ENET_STATUS_TX_BUFF_BUSY	All internal TX buffers are currently in use
ENET_STATUS_TX_BUFF_OVERFLOW	The requested TX buffer length cannot

Definition at line 107 of file Enet_Ip_Types.h.

7.2.4.2 Enet_Ip_EventType

enum `Enet_Ip_EventType`

Event specifier for the callback function.

Definition at line 156 of file `Enet_Ip_Types.h`.

7.2.4.3 Enet_Ip_TxSpecialConfigType

enum `Enet_Ip_TxSpecialConfigType`

Special transmit control configurations.

Definition at line 184 of file `Enet_Ip_Types.h`.

7.2.4.4 Enet_Ip_TxAcceleratorType

enum `Enet_Ip_TxAcceleratorType`

Transmit accelerator configurations.

Definition at line 194 of file `Enet_Ip_Types.h`.

7.2.4.5 Enet_Ip_RxAcceleratorType

enum `Enet_Ip_RxAcceleratorType`

Receive accelerator configurations.

Definition at line 205 of file `Enet_Ip_Types.h`.

7.2.4.6 Enet_Ip_CounterType

enum `Enet_Ip_CounterType`

Statistics counters enumeration.

Definition at line 238 of file `Enet_Ip_Types.h`.

7.2.4.7 Enet_Ip_MiiSpeedType

enum `Enet_Ip_MiiSpeedType`

Media Independent Interface speed selection.

Enumerator

<code>ENET_MII_SPEED_10M</code>	Speed 10 Mbps.
<code>ENET_MII_SPEED_100M</code>	Speed 100 Mbps.

Definition at line 301 of file `Enet_Ip_Types.h`.

7.2.4.8 Enet_Ip_PowerStateType

enum `Enet_Ip_PowerStateType`

Controller power state indication.

Enumerator

ENET_PSTATE_INACTIVE	Controller is powered on and Transmitter/Receiver are disabled
ENET_PSTATE_ACTIVE	Controller is powered on and Transmitter/Receiver are enabled
ENET_PSTATE_SLEEP	Controller is sleeping (WoL or LPI) and waiting for wake-up packet
ENET_PSTATE_RESET	Controller is resetting its internal logic and registers

Definition at line 333 of file `Enet_Ip_Types.h`.

7.2.4.9 Enet_Ip_SysTimeCorrOffsetType

enum `Enet_Ip_SysTimeCorrOffsetType`

System time correction offset.

Enumerator

ENET_SYS_TIME_CORR_POSITIVE	Correction value is added to the system time.
ENET_SYS_TIME_CORR_NEGATIVE	Correction value is subtracted from the system time.

Definition at line 434 of file `Enet_Ip_Types.h`.

7.2.4.10 Enet_Ip_TimerChannelModeType

enum `Enet_Ip_TimerChannelModeType`

Timer channels configuration modes.

Enumerator

ENET_TIMER_OC_SOFTWARE	Timer Channel is configured for Output Compare - software only.
------------------------	---

Definition at line 493 of file `Enet_Ip_Types.h`.

7.2.4.11 Enet_Ip_ChecksumInsControlType

enum `Enet_Ip_ChecksumInsControlType`

Checksum insertion control.

Enumerator

ENET_CHECKSUM_INSERTION_DISABLE	Checksum insertion is disabled.
ENET_CHECKSUM_INSERTION_IP	IP header checksum calculation and insertion are enabled.
ENET_CHECKSUM_INSERTION_PROTO	Insert protocol specific checksum.
ENET_CHECKSUM_INSERTION_IP_PROTO	Both Ip header and protocol are enabled

Definition at line 529 of file Enet_Ip_Types.h.

7.2.4.12 Enet_Ip_MmfrOpType

enum [Enet_Ip_MmfrOpType](#)

Management Frame operation type.

Definition at line 540 of file Enet_Ip_Types.h.

7.2.4.13 Enet_Ip_FifoThresholdType

enum [Enet_Ip_FifoThresholdType](#)

Available FIFO threshold levels.

Enumerator

ENET_RX_SECTION_FULL	Value, in 64-bit words, of the receive FIFO section full threshold. When the FIFO level reaches the value in this field, data is available in the Receive FIFO.
ENET_RX_SECTION_EMPTY	Value, in 64-bit words, of the receive FIFO section empty threshold. When the FIFO has reached this level, a pause frame will be issued.
ENET_RX_ALMOST_EMPTY	Value, in 64-bit words, of the receive FIFO almost empty threshold. When the FIFO level reaches this level and the end-of-frame has not been received for the frame yet, the core receive read control stops FIFO read. A minimum value of 4 should be set.
ENET_RX_ALMOST_FULL	Value, in 64-bit words, of the receive FIFO almost full threshold. When the FIFO level comes close to the maximum, so that there is no more space for at least RX_ALMOST_FULL number of words, the MAC stops writing data in the FIFO and truncates the received frame to avoid FIFO overflow. A minimum value of 4 should be set.
ENET_TX_SECTION_EMPTY	Value, in 64-bit words, of the transmit FIFO section empty threshold. When the FIFO level reaches this value, a MAC status signal indicates that the transmit FIFO is getting full. This gives the ENET module an indication to slow or stop its write transaction to avoid a buffer overflow.
ENET_TX_ALMOST_EMPTY	Value, in 64-bit words, of the transmit FIFO almost empty threshold. When the FIFO level reaches this level and no end-of-frame is available for the frame, the MAC transmit logic, to avoid FIFO underflow, stops reading the FIFO and transmits a frame with an MII error indication.

Enumerator

ENET_TX_ALMOST_FULL	Value, in 64-bit words, of the transmit FIFO almost full threshold. A minimum value of six is required. When the FIFO level comes close to the maximum, so that there is no more space for at least TX_ALMOST_FULL number of words, the FIFO write control logic, to avoid FIFO overflow, truncates the current frame and sets the error status. As a result, the frame will be transmitted with an GMII/MII error indication.
---------------------	--

Definition at line 562 of file Enet_Ip_Types.h.

7.2.5 Function Reference

7.2.5.1 Enet_Ip_Init()

```
void Enet_Ip_Init (
    uint8 Instance,
    const Enet_CtrlConfigType * Config )
```

Initializes the ENET module.

This function initializes and enables the ENET module, configuring receive and transmit control settings, the receive and transmit descriptors rings, and the MAC physical address.

Note: All untagged packets will be received by ring 0 (zero). VLAN tagged packets are routed depending on the VLAN Tag Priority field according to the provided configuration.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Config</i>	The module configuration structure

7.2.5.2 Enet_Ip_GetPowerState()

```
Enet_Ip_PowerStateType Enet_Ip_GetPowerState (
    uint8 Instance )
```

Gets the current power state of the ENET module.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

Returns

Enet_Ip_PowerStateType The power state of the controller

7.2.5.3 Enet_Ip_Deinit()

```
void Enet_Ip_Deinit (
    uint8 Instance )
```

Deinitializes the ENET module.

This function disables the interrupts and then disables the ENET module.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.4 Enet_Ip_EnableController()

```
void Enet_Ip_EnableController (
    uint8 Instance )
```

Enables all configured transmit and receive buffers and then enables the controller.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.5 Enet_Ip_RecoverFilter()

```
void Enet_Ip_RecoverFilter (
    uint8 Instance )
```

Recover the hash table filtering caused by add add/remove open/close to hash table.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.6 Enet_Ip_DisableController()

```
void Enet_Ip_DisableController (
    uint8 Instance )
```

Disables the controller and resets all the configured transmit and receive buffers.

Warning: This function ignores all pending transmission and reception requests

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.7 Enet_Ip_SetSpeed()

```
void Enet_Ip_SetSpeed (
    uint8 Instance,
    Enet_Ip_MiiSpeedType Speed )
```

Sets the speed of the MII interface.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Speed</i>	MII speed

7.2.5.8 Enet_Ip_GetTxBuff()

```
Enet_Ip_StatusType Enet_Ip_GetTxBuff (
    uint8 Instance,
    uint8 Ring,
    Enet_Ip_BufferType * Buff,
    uint16 * BuffId )
```

Provides a transmit buffer to be used by the application for transmission.

This function provides an internal buffer which can further be used by the application to store the transmit data.

Note: The buffer will be marked as locked and won't be released until after a call to Enet_Ip_GetTransmitStatus for the same buffer returns ENET_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to get a buffer at the same time. This synchronization shall be implemented by the application.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Ring</i>	Ring number
in, out	<i>Buff</i>	In: Buffer containing the desired length Out: Buffer containing the granted length or available length in case of overflow.
out	<i>BuffId</i>	Index of the buffer (descriptor) within the ring. If this information is not needed, this parameter should be NULL_PTR.

Return values

<i>ENET_STATUS_SUCCESS</i>	The buffer has been successfully locked.
<i>ENET_STATUS_TX_BUFF_BUSY</i>	All buffers are currently in use.
<i>ENET_STATUS_TX_BUFF_OVERFLOW</i>	The requested buffer length cannot be granted.

7.2.5.9 Enet_Ip_SendFrame()

```
Enet_Ip_StatusType Enet_Ip_SendFrame (
    uint8 Instance,
    uint8 Ring,
    const Enet_Ip_BufferType * Buff,
    const Enet_Ip_TxOptionsType * Options )
```

Sends an Ethernet frame.

This function sends an Ethernet frame, contained in the buffer received as parameter.

Note: Since the transmission of the frame is not complete when this function returns, the application must not change/alter/re-use the provided buffer until after a call to Enet_Ip_GetTransmitStatus for the same buffer returns ENET_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to send a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Ring</i>	The ring number
in	<i>Buff</i>	The buffer containing the frame
in	<i>Options</i>	Configuration options applicable to this buffer's transmission only. Can be NULL_PTR, if no special option is required.

Return values

<i>ENET_STATUS_SUCCESS</i>	The frame was successfully enqueued for transmission.
<i>ENET_STATUS_TX_QUEUE_FULL</i>	There is no available space for the frame in the queue.

7.2.5.10 Enet_Ip_ReadFrame()

```
Enet_Ip_StatusType Enet_Ip_ReadFrame (
    uint8 Instance,
    uint8 Ring,
    Enet_Ip_BufferType * Buff,
    Enet_Ip_RxInfoType * Info )
```

Reads a received Ethernet frame.

This function reads the first received Ethernet frame in the Rx queue. The buffer received as parameter will be updated by the driver and the .data field will point to a memory area containing the frame data.

Note: Once the application finished processing the buffer, it could be reused by the driver for further receptions by invoking Enet_Ip_ProvideRxBuff.

Important: The driver does not ensure synchronization between different threads trying to read a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Ring</i>	The ring number
out	<i>Buff</i>	The buffer containing the frame
out	<i>Info</i>	Enhanced information related to the data contained by this receive buffer. If this information is not needed, this parameter should be NULL_PTR.

Return values

<i>ENET_STATUS_SUCCESS</i>	A frame was successfully read.
<i>ENET_STATUS_RX_QUEUE_EMPTY</i>	There is no available frame in the queue.

7.2.5.11 Enet_Ip_ProvideRxBuff()

```
void Enet_Ip_ProvideRxBuff (
    uint8 Instance,
    uint8 Ring,
    const Enet_Ip_BufferType * Buff )
```

Provides a receive buffer to be used by the driver for reception.

This function provides a buffer which can further be used by the reception mechanism in order to store the received data.

Note: The application can either provide a buffer previously obtained in a `Enet_Ip_ReadFrame` call (when it is no longer needed after being fully processed), or allocate a new buffer. The former approach is recommended as it has a simpler usage model and re-uses the same initial memory range for the entire driver lifetime operation. The later approach could provide more flexibility, but since it involves constant memory free/alloc operations it is only recommended with an efficient pool-based memory allocator.

Important: The driver does not ensure synchronization between different threads trying to provide a buffer at the same time. This synchronization shall be implemented by the application.

Important: The application is responsible for providing one Rx buffer for every frame it receives, otherwise the reception ring can fill-up, affecting further reception.

Usage example:

```
stat = Enet_Ip_ReadFrame(INST_ENET0, 0U, &rxBuff);
if (stat == ENET_STATUS_SUCCESS) { process_buffer(&rxBuff); Enet_Ip_ProvideRxBuff(INST_ENET0, 0U,
&rxBuff); }
```

Parameters

in	<i>Instance</i>	Instance number
in	<i>Ring</i>	The ring number
in	<i>Buff</i>	The buffer to be added to the reception ring

7.2.5.12 Enet_Ip_IsFrameAvailable()

```
boolean Enet_Ip_IsFrameAvailable (
    uint8 Instance,
    uint8 Ring )
```

Checks if there are more frames available in the given queue.

This function takes a peek at the given Rx queue to check if there are more Ethernet frames to be received. Its intended usage is to provide this information without also extracting the frame as "Enet_Ip_ReadFrame".

Parameters

in	<i>Instance</i>	Instance number
in	<i>Ring</i>	The ring number

Return values

<i>TRUE</i>	There is an available frame in the queue.
<i>FALSE</i>	There is no available frame in the queue.

7.2.5.13 Enet_Ip_GetTransmitStatus()

```
Enet_Ip_StatusType Enet_Ip_GetTransmitStatus (
    uint8 Instance,
    uint8 Ring,
    const Enet_Ip_BufferType * Buff,
    Enet_Ip_TxInfoType * Info )
```

Checks if the transmission of a buffer is complete.

This function checks if the transmission of the given buffer is complete.

Note: If the buffer is not found in the Tx ring, the function will return ENET_STATUS_BUFF_NOT_FOUND.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Ring</i>	The ring number
in	<i>Buff</i>	The transmit buffer for which the status shall be checked
out	<i>Info</i>	Extended information related to the buffer. If this information is not needed, this parameter should be NULL_PTR.

Return values

<i>ENET_STATUS_BUSY</i>	The frame is still enqueued for transmission.
<i>ENET_STATUS_BUFF_NOT_FOUND</i>	The buffer was not found in the Tx queue.
<i>ENET_STATUS_SUCCESS</i>	Otherwise.

7.2.5.14 Enet_Ip_GetCounter()

```
uint32 Enet_Ip_GetCounter (
    uint8 Instance,
    Enet_Ip_CounterType Counter )
```

Gets statistics from the specified counter.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Counter</i>	The counter to be read

Returns

The value of the requested counter

7.2.5.15 Enet_Ip_ConfigCounters()

```
void Enet_Ip_ConfigCounters (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables the MIB counters.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Enable</i>	The MIB logic is enabled(TRUE) else disabled(FALSE)

Returns

The Value of the requested counter

7.2.5.16 Enet_Ip_EnableMDIO()

```
void Enet_Ip_EnableMDIO (
    uint8 Instance,
    boolean MiiPreambleDisabled,
    uint32 ModuleClk )
```

Enables the MDIO interface.

Parameters

in	<i>Instance</i>	Instance number
in	<i>MiiPreambleDisabled</i>	Enables/disables prepending a preamble to the MII management frame.
in	<i>ModuleClk</i>	Clock frequency provide MDC.

7.2.5.17 Enet_Ip_MDIORead()

```
Enet_Ip_StatusType Enet_Ip_MDIORead (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 PhyReg,
    uint16 * Data,
    uint32 TimeoutMs )
```

Reads the selected register of the PHY.

Parameters

in	<i>Instance</i>	Instance number
in	<i>PhyAddr</i>	PHY device address
in	<i>PhyReg</i>	PHY register address
out	<i>Data</i>	Data read from the PHY
in	<i>TimeoutMs</i>	Timeout for the read operation (in milliseconds)

Return values

<i>ENET_STATUS_SUCCESS</i>	The operation completed successfully.
<i>ENET_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.5.18 Enet_Ip_MDIOWrite()

```
Enet_Ip_StatusType Enet_Ip_MDIOWrite (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 PhyReg,
    uint16 Data,
    uint32 TimeoutMs )
```

Writes the selected register of the PHY.

Parameters

in	<i>Instance</i>	Instance number
in	<i>PhyAddr</i>	PHY device address
in	<i>PhyReg</i>	PHY register address
in	<i>Data</i>	Data to be written in the specified register of the PHY
in	<i>TimeoutMs</i>	Timeout for the write operation (in milliseconds)

Return values

<i>ENET_STATUS_SUCCESS</i>	The operation completed successfully.
<i>ENET_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.5.19 Enet_Ip_MDIOReadMMD()

```
Enet_Ip_StatusType Enet_Ip_MDIOReadMMD (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 Mmd,
    uint16 PhyReg,
    uint16 * Data,
    uint32 TimeoutMs )
```

Reads a register of the specified MMD in a PHY device.

Parameters

in	<i>Instance</i>	Instance number
in	<i>PhyAddr</i>	PHY device address
in	<i>Mmd</i>	The MMD index of the target register
in	<i>PhyReg</i>	PHY register address
out	<i>Data</i>	Data read from the PHY
in	<i>TimeoutMs</i>	Timeout for the read operation (in milliseconds)

Return values

<i>ENET_STATUS_SUCCESS</i>	The operation completed successfully.
<i>ENET_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.5.20 Enet_Ip_MDIOWriteMMD()

```
Enet_Ip_StatusType Enet_Ip_MDIOWriteMMD (
    uint8 Instance,
    uint8 PhyAddr,
    uint8 Mmd,
    uint16 PhyReg,
    uint16 Data,
    uint32 TimeoutMs )
```

Writes a register of the specified MMD in a PHY device.

Parameters

in	<i>Instance</i>	Instance number
in	<i>PhyAddr</i>	PHY device address
in	<i>Mmd</i>	The MMD index of the target register
in	<i>PhyReg</i>	PHY register address
in	<i>Data</i>	Data to be written in the specified register of the PHY
in	<i>TimeoutMs</i>	Timeout for the write operation (in milliseconds)

Return values

<i>ENET_STATUS_SUCCESS</i>	The operation completed successfully.
<i>ENET_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.5.21 Enet_Ip_SetMacAddr()

```
void Enet_Ip_SetMacAddr (
    uint8 Instance,
    const uint8 * MacAddr )
```

Configures the physical address of the MAC.

Parameters

in	<i>Instance</i>	Instance number
in	<i>MacAddr</i>	The MAC address to be configured

7.2.5.22 Enet_Ip_GetMacAddr()

```
void Enet_Ip_GetMacAddr (
    uint8 Instance,
    uint8 * MacAddr )
```

Gets the physical address of the MAC.

Parameters

in	<i>Instance</i>	Instance number
out	<i>MacAddr</i>	The physical address of the MAC

7.2.5.23 Enet_Ip_SetBroadcastForwardAll()

```
void Enet_Ip_SetBroadcastForwardAll (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables forwarding of the broadcast traffic.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Enable</i>	If true, the application will receive all the broadcast traffic; if false, it stops forwarding this kind of traffic.

7.2.5.24 Enet_Ip_SetMulticastForwardAll()

```
void Enet_Ip_SetMulticastForwardAll (
    uint8 Instance,
    boolean Enable )
```

Enables/Disables forwarding of the multicast traffic, irrespective of the destination MAC address.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Enable</i>	If true, the application will receive all the multicast traffic; if false, it stops forwarding this kind of traffic.

7.2.5.25 Enet_Ip_AddDstAddrToHashFilter()

```
void Enet_Ip_AddDstAddrToHashFilter (
    uint8 Instance,
    const uint8 * MacAddr )
```

Adds a hardware address to the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

Parameters

in	<i>Instance</i>	Instance number
in	<i>MacAddr</i>	The physical address

7.2.5.26 Enet_Ip_RemoveDstAddrFromHashFilter()

```
void Enet_Ip_RemoveDstAddrFromHashFilter (
    uint8 Instance,
    const uint8 * MacAddr )
```

Removes a hardware address from the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

Parameters

in	<i>Instance</i>	Instance number
in	<i>MacAddr</i>	The physical address

7.2.5.27 Enet_Ip_SetSysTimeCorr()

```
void Enet_Ip_SetSysTimeCorr (
    uint8 Instance,
    Enet_Ip_SysTimeCorrOffsetType Offset,
    Enet_Ip_TimeStampType DiffTimeStamp )
```

Set system time correction.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Offset</i>	Chooses between negative and positive correction
in	<i>DiffTimeStamp</i>	Offset between time stamp grandmaster and time stamp by local clock.

7.2.5.28 Enet_Ip_TimerSetCorrection()

```
void Enet_Ip_TimerSetCorrection (
    uint8 Instance,
    uint32 CorrectValue,
    uint32 CorrectPeriod )
```

Sets the timer correction period and correction increment value.

Parameters

in	<i>Instance</i>	Instance number
in	<i>CorrectValue</i>	Correction timer added.
in	<i>CorrectPeriod</i>	Number of timer clock cycles

7.2.5.29 Enet_Ip_TimerStop()

```
void Enet_Ip_TimerStop (
    uint8 Instance )
```

The timer stop at the current value.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.30 Enet_Ip_TimerEnableChannel()

```
Enet_Ip_StatusType Enet_Ip_TimerEnableChannel (
    uint8 Instance,
    uint8 Channel,
    const Enet_Ip_TimerChannelConfigType * Config )
```

Configures and enables a specific timer channel.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Channel</i>	Timer channel

Parameters

in	<i>Config</i>	the value(mode, interrupt, enable DMA or not) to set for timer channel.
----	---------------	---

Return values

<i>ENET_STATUS_SUCCESS</i>	The operation completed successfully.
<i>ENET_STATUS_TIMEOUT</i>	The specified timeout expired before completing the operation.

7.2.5.31 Enet_Ip_TimerSetCompare()

```
void Enet_Ip_TimerSetCompare (
    uint8 Instance,
    uint8 Channel,
    uint32 Value )
```

Sets the compare value for a channel configured as output compare.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Channel</i>	Timer channel
in	<i>Value</i>	the value to compare.

7.2.5.32 Enet_Ip_TimerStart()

```
void Enet_Ip_TimerStart (
    uint8 Instance )
```

The timer starts incrementing.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.33 Enet_Ip_TimerInit()

```
void Enet_Ip_TimerInit (
    uint8 Instance,
    const Enet_Ip_TimerConfigType * TimerConfig )
```

Initializes the adjustable timer using the given configuration and enables it.

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

Parameters

in	<i>TimerConfig</i>	timer value need to initialize.
----	--------------------	---------------------------------

7.2.5.34 Enet_Ip_TimerSet()

```
void Enet_Ip_TimerSet (
    uint8 Instance,
    Enet_Ip_TimeStampType TimerValue )
```

Sets the value of the timer.

Parameters

in	<i>Instance</i>	Instance number
in	<i>TimerConfig</i>	timer value set.

7.2.5.35 Enet_Ip_TimerGet()

```
void Enet_Ip_TimerGet (
    uint8 Instance,
    Enet_Ip_TimeStampType * TimeStamp )
```

Gets the value of the timer.

Parameters

in	<i>Instance</i>	Instance number
in	<i>TimeStamp</i>	timer value gotten.

7.2.5.36 Enet_Ip_EnableTxStoreAndForward()

```
void Enet_Ip_EnableTxStoreAndForward (
    uint8 Instance )
```

Enable store and forward .

Parameters

in	<i>Instance</i>	Instance number
----	-----------------	-----------------

7.2.5.37 Enet_Ip_SetFIFOThreshold()

```
void Enet_Ip_SetFIFOThreshold (
    uint8 Instance,
```

```
Enet_Ip_FifoThresholdType ThresholdType,
uint8 ThresholdValue )
```

Sets FIFO threshold levels.

Parameters

in	<i>Instance</i>	Instance number
in	<i>ThresholdType</i>	FIFO threshold types (defined in Enet_Ip_FifoThresholdType) need to update
in	<i>ThresholdValue</i>	FIFO threshold value

7.2.5.38 Enet_Ip_SetTxFIFOWatermark()

```
void Enet_Ip_SetTxFIFOWatermark (
    uint8 Instance,
    uint16 Watermark )
```

Sets the transmit FIFO watermark.

Parameters

in	<i>Instance</i>	Instance number
in	<i>Watermark</i>	Number of bytes written to the transmit FIFO before transmission.

7.3 Ethernet Driver

7.3.1 Detailed Description

Macros

- #define [ETH_43_ENET_E_COMMUNICATION](#)
Runtime Error ID for "Failure or incorrect communication with the Ethernet Controller".

Function Reference

- void [Eth_43_ENET_Init](#) (const Eth_43_ENET_ConfigType *CfgPtr)
Initializes the Ethernet Driver.
- Std_ReturnType [Eth_43_ENET_SetControllerMode](#) (uint8 CtrlIdx, Eth_ModeType CtrlMode)
Enables or disables the given controller.
- Std_ReturnType [Eth_43_ENET_GetControllerMode](#) (uint8 CtrlIdx, Eth_ModeType *CtrlModePtr)
Obtains the mode of the given controller.
- Std_ReturnType [Eth_43_ENET_TxTimeAwareShaperInit](#) (uint8 CtrlIdx)
Control transmit frame in duration time (This function is obsolete).
- void [Eth_43_ENET_GetPhysAddr](#) (uint8 CtrlIdx, uint8 *PhysAddrPtr)
Obtains the physical source address used by the indexed controller (the node MAC address).
- void [Eth_43_ENET_SetPhysAddr](#) (uint8 CtrlIdx, const uint8 *PhysAddrPtr)
Set or change physical address to the defined controller.
- BufReq_ReturnType [Eth_43_ENET_ProvideTxBuffer](#) (uint8 CtrlIdx, uint8 Priority, Eth_BufIdxType *BufIdxPtr, uint8 **BufPtr, uint16 *LenBytePtr)
Provides access to a transmit buffer of the specified controller.
- Std_ReturnType [Eth_43_ENET_Transmit](#) (uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_FrameType FrameType, boolean TxConfirmation, uint16 LenByte, const uint8 *PhysAddrPtr)
Triggers transmission of a previously granted and then filled transmit buffer.
- void [Eth_43_ENET_Receive](#) (uint8 CtrlIdx, uint8 FifoIdx, Eth_RxStatusType *RxStatusPtr)
Triggers frames reception notifications.
- void [Eth_43_ENET_TxConfirmation](#) (uint8 CtrlIdx)
Triggers frame transmission confirmations.

7.3.2 Macro Definition Documentation

7.3.2.1 ETH_43_ENET_E_COMMUNICATION

```
#define ETH_43_ENET_E_COMMUNICATION
```

Runtime Error ID for "Failure or incorrect communication with the Ethernet Controller".

Definition at line 306 of file Eth_43_ENET.h.

7.3.3 Function Reference

7.3.3.1 Eth_43_ENET_Init()

```
void Eth_43_ENET_Init (
    const Eth_43_ENET_ConfigType * CfgPtr )
```

Initializes the Ethernet Driver.

The configuration pointer is internally stored and the driver is initialized. The Ethernet controller is also reset.

Note

Function should be called only once.

Warning

Second call can cause undefined behavior. Call the Eth_SetControllerMode() and pass ETH_MODE_DOWN to the CtrlMode argument before the second Eth_Init call to avoid problems.

Parameters

in	<i>CfgPtr</i>	Points to the implementation specific structure containing the Eth driver configuration Compiler_Warning: this warning due to behavior of compiler depend on configs. implements Eth_Init_Activity
----	---------------	---

7.3.3.2 Eth_43_ENET_SetControllerMode()

```
Std_ReturnType Eth_43_ENET_SetControllerMode (
    uint8 CtrlIdx,
    Eth_ModeType CtrlMode )
```

Enables or disables the given controller.

Warning

Disabling the controller clears all receive and transmit buffers. The application should ensure that no data is lost.

Parameters

in	<i>CtrlIdx</i>	Index of the controller to be enabled or disabled. The index is valid within the context of the Ethernet Driver only.
in	<i>CtrlMode</i>	Mode which shall be entered - ETH_MODE_DOWN: disable the controller - ETH_MODE_ACTIVE: enable the controller

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed. implements Eth_SetControllerMode_Activity

7.3.3.3 Eth_43_ENET_GetControllerMode()

```
Std_ReturnType Eth_43_ENET_GetControllerMode (
    uint8 CtrlIdx,
    Eth_ModeType * CtrlModePtr )
```

Obtains the mode of the given controller.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.
out	<i>CtrlModePtr</i>	Pointer where to store the current controller mode.

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed. implements Eth_GetControllerMode_Activity

7.3.3.4 Eth_43_ENET_TxTimeAwareShaperInit()

```
Std_ReturnType Eth_43_ENET_TxTimeAwareShaperInit (
    uint8 CtrlIdx )
```

Control transmit frame in duration time (This function is obsolete).

Parameters

in	<i>CtrlIdx</i>	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.
----	----------------	---

Warning

This function is obsolete. If time aware shaper feature is intended to be used, call Eth_StartTas and Eth_StopTas instead.

Returns

Std_ReturnType

7.3.3.5 Eth_43_ENET_GetPhysAddr()

```
void Eth_43_ENET_GetPhysAddr (
    uint8 CtrlIdx,
    uint8 * PhysAddrPtr )
```

Obtains the physical source address used by the indexed controller (the node MAC address).

Parameters

in	<i>CtrlIdx</i>	Index of the controller whose MAC address should be read. The index is valid within the context of the Ethernet Driver only.
out	<i>PhysAddrPtr</i>	Pointer where to store the physical source address (MAC address). The address in network byte order is stored into 6 bytes at the given memory address. implements Eth_GetPhysAddr_Activity

7.3.3.6 Eth_43_ENET_SetPhysAddr()

```
void Eth_43_ENET_SetPhysAddr (
    uint8 CtrlIdx,
    const uint8 * PhysAddrPtr )
```

Set or change physical address to the defined controller.

Parameters

in	<i>CtrlIdx</i>	Index of the controller whose MAC address should be changed. The index is valid within the context of the Ethernet Driver only.
in	<i>PhysAddrPtr</i>	Pointer to the physical source address which should be set to the controller. The address is stored in 6 bytes of memory in network byte order.

Warning

This function may be called only when the controller is down. Calling function Eth_Init will change the controller's MAC address to the default value! implements Eth_SetPhysAddr_Activity

7.3.3.7 Eth_43_ENET_ProvideTxBuffer()

```
BufReq_ReturnType Eth_43_ENET_ProvideTxBuffer (
    uint8 CtrlIdx,
    uint8 Priority,
    Eth_BufIdxType * BufIdxPtr,
    uint8 ** BufPtr,
    uint16 * LenBytePtr )
```

Provides access to a transmit buffer of the specified controller.

Warning

The application should handle possible difference between the requested and granted buffer lengths. It is not necessary to use whole granted buffer i.e. some space at the end may not be written.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which buffer shall be provided. The index is valid within the context of the Ethernet Driver only.
in	<i>Priority</i>	Frame priority for transmit buffer FIFO selection

Parameters

out	<i>BufIdxPtr</i>	Index to the granted transmit buffer resource. It uniquely identifies the buffer in all subsequent calls of functions <code>Eth_Transmit()</code> and <code>Eth_TxConfirmation()</code> .
out	<i>BufPtr</i>	Pointer to the granted buffer. This is the space where the data to be transmitted shall be stored.
in, out	<i>LenBytePtr</i>	Buffer payload length - In: desired length in bytes - Out: granted length in bytes

Returns

Error and buffer status

Return values

<i>BUFREQ_OK</i>	Buffer was successfully granted and no error has occurred.
<i>BUFREQ_E_NOT_OK</i>	A development error was detected and no buffer was granted.
<i>BUFREQ_E_BUSY</i>	All available buffers in use therefore no buffer was granted. No error has been detected. implements <code>Eth_ProvideTxBuffer_Activity</code>

7.3.3.8 Eth_43_ENET_Transmit()

```
Std_ReturnType Eth_43_ENET_Transmit (
    uint8 CtrlIdx,
    Eth_BufIdxType BufIdx,
    Eth_FrameType FrameType,
    boolean TxConfirmation,
    uint16 LenByte,
    const uint8 * PhysAddrPtr )
```

Triggers transmission of a previously granted and then filled transmit buffer.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which buffer shall be transmitted. The index is valid within the context of the Ethernet Driver only.
in	<i>BufIdx</i>	Index of the buffer resource to be transmitted.
in	<i>FrameType</i>	Desired value of the Ethernet frame type in the frame header.
in	<i>TxConfirmation</i>	Activates transmission confirmation.
in	<i>LenByte</i>	Buffer data length in bytes (payload length).
in	<i>PhysAddrPtr</i>	Physical target address (MAC address) in network byte order.

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
-------------	--

Return values

<i>E_NOT_OK</i>	Development error was detected and the function failed. implements Eth_Transmit_Activity
-----------------	--

7.3.3.9 Eth_43_ENET_Receive()

```
void Eth_43_ENET_Receive (
    uint8 CtrlIdx,
    uint8 FifoIdx,
    Eth_RxStatusType * RxStatusPtr )
```

Triggers frames reception notifications.

All receive buffers are checked and the first received frame is passed to the EthIf module. The caller is notified whether any frame was received and whether more frames are available in the receive queue.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which shall be checked whether any new frames were received. The index is valid within the context of the Ethernet Driver only.
in	<i>FifoIdx</i>	Specifies the related fifo
out	<i>RxStatusPtr</i>	Informs the caller whether a frame was received (ETH_RECEIVED or ETH_NOT_RECEIVED) and whether more frames are available in the queue (ETH_RECEIVED or ETH_RECEIVED_MORE_DATA_AVAILABLE). implements Eth_Receive_Activity

7.3.3.10 Eth_43_ENET_TxConfirmation()

```
void Eth_43_ENET_TxConfirmation (
    uint8 CtrlIdx )
```

Triggers frame transmission confirmations.

Parameters

in	<i>CtrlIdx</i>	Index of the controller which shall be checked whether any frame transmission has finished. The index is valid within the context of the Ethernet Driver only.
----	----------------	--

All transmit buffers are checked and upper layers are informed about successfully transmitted frames. Buffers containing transmitted frames are unlocked after the confirmation. implements Eth_TxConfirmation_Activity

Chapter 8

Data Structure Documentation

8.1 Enet_Ip_ManagementInfo Struct Reference

Management Frame information.

```
#include <Enet_Ip_Hw_Access.h>
```

8.1.1 Detailed Description

Management Frame information.

Definition at line 139 of file Enet_Ip_Hw_Access.h.

How to Reach Us:

Home Page:

nxp.com

Web Support:

nxp.com/support

Limited warranty and liability - Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes - NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use in automotive applications - This NXP product has been qualified for use in automotive applications. If this product is used by customer in the development of, or for incorporation into, products or services (a) used in safety critical applications or (b) in which failure could lead to death, personal injury, or severe physical or environmental damage (such products and services hereinafter referred to as "Critical Applications"), then customer makes the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, safety, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP. As such, customer assumes all risk related to use of any products in Critical Applications and NXP and its suppliers shall not be liable for any such use by customer. Accordingly, customer will indemnify and hold NXP harmless from any claims, liabilities, damages and associated costs and expenses (including attorneys' fees) that NXP may incur related to customer's incorporation of any product in a Critical Application.

Applications - Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale - NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at <http://www.nxp.com/profile/terms>, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Export control - This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at <mailto:PSIRT@nxp.com>) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

NXP B.V. - NXP B.V. is not an operating company and it does not distribute or sell products.